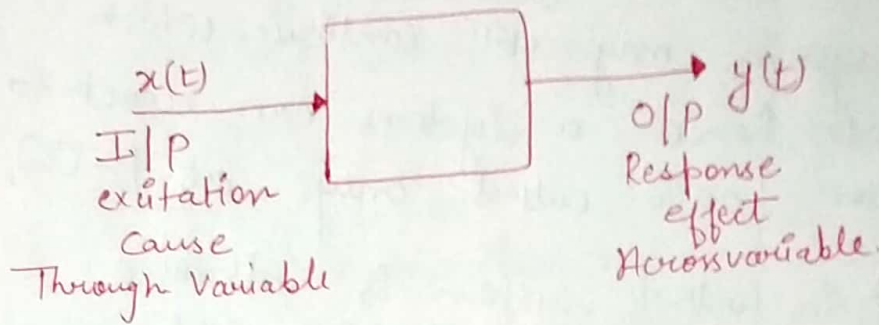


CONTROL SYSTEM

"Means of transforming a signal"
 $\rightarrow f(t, \dots)$
 $\hookrightarrow$ energy

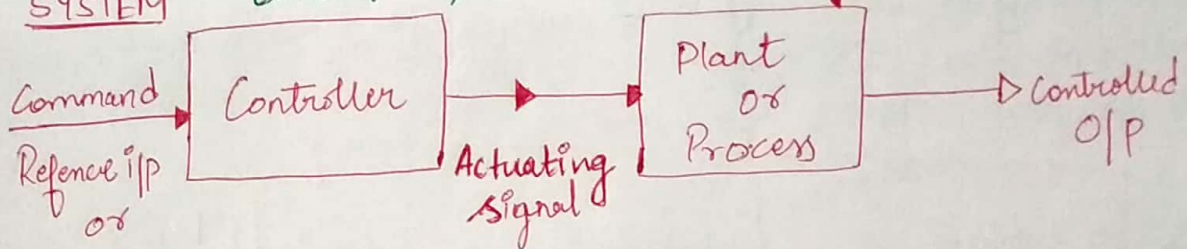


$$v \propto i$$

$$v = Ri$$

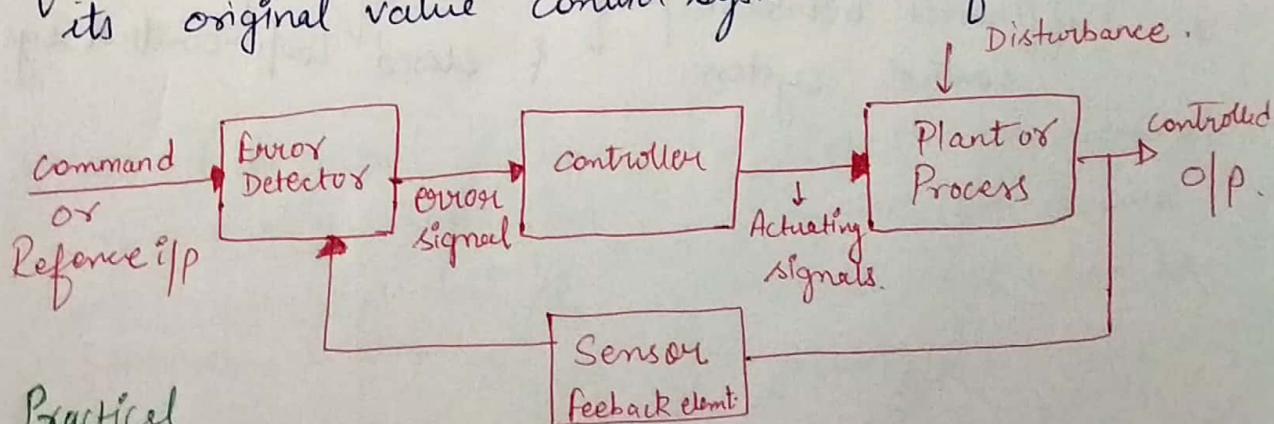
* Control System: is that means by which any quantity of interest is maintained or altered according to the desired manner.

* IDEAL CONTROL SYSTEM Control System has atleast two sub-systems.



Desired o/p

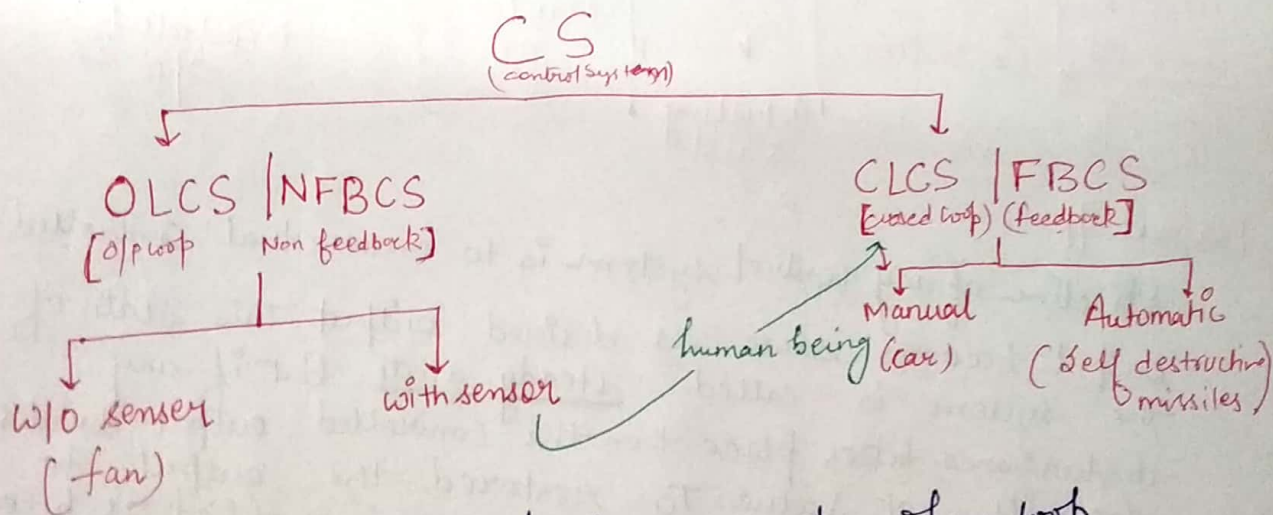
Objective of any control system is to ensure that Controlled output becomes same as desired output. This state of the system is called steady state. But if any disturbance takes place then the controlled output deviates from its set value. To restore the output to its original value control system is modified as below:



Practical control System

This design can overcome the effect of disturbance associated with plant of the system in the output of system whereas disturbance with other parts of the system may still continue, which can't be avoided hence a system can reach to steady state with 100% output only at $t = \infty$.

Though feedback in control system is introduced mainly to improve its accuracy, it also has effect on the several other characteristics of the system like gain, Bandwidth, Speed, Stability, Sensitivity etc.



* Differences between performance of open loop control system & closed loop control system.

OPEN LOOP

- ① The Behaviour of open loop system doesn't vary though its o/p varies. Hence open loop control system is not accurate.
- ② Open loop system has not complete sense but may have sensor.
- ③ Time constant of open loop system is larger due to which transients die out slowly. Hence open loop system is slow.
- ④ The affect of external disturbance and internal parameter variation is more in o/p loop. i.e open loop system is more sensitivity.
- ⑤ Open loop system is simple and economical.
- ⑥ Open loop control system is generally stable but can't stabilize but if becomes unstable.

CLOSED LOOP

- The behaviour of the closed loop system does change if its output varies due to any disturbance. Hence closed loop system is accurate.
- closed loop control system has complete sense either manually or automatically.
 - Time constant of closed loop system is smaller due to which transient dies out rapidly hence closed loop system is faster.
 - The affect of external disturbance and internal parameter variation is less in close loop. i.e CLS is less sensitive.
 - closed loop system is complex and expensive.
 - closed loop control system can become unstable but can be stabilized.

1. No feedback guarantees stability or instability, the $-ive$ feedback always guarantees ^{more stability} than positive feedback.
2. Inspite of having $-ive$ feedback control system can still become unstable due to high open loop gain, high type number, high sensitivity, high transportation delay lag phase.

* Differences b/w performances of negative feedback closed loop system & $+ive$ feedback closed loop system

Performance Criterion	$-ive$ f.B	$+ive$ f.B.
Gain	↓ decreases.	↑
Bandwidth	↑ increases	↓
Time Constant	↓	↑
Speed	↑	↓
Sensitivity	↓	↑
Stability	↑	↓

"By default control system is closed loop negative feedback."

	T.D	f.D
C.T	L.T	Z.T
x D.T	Z.T	DTFT

Mathematical Models

- 1). Transfer funⁿ (LTI)
- 2). State model (Any)

$$\text{Total Response} = \underset{\substack{\text{(Due to} \\ \text{"initial cond"')}}}{ZIR} + \underset{\substack{\text{(Due to } i|P)}}{ZCSR}$$

$$\text{Transfer fun}^n = \frac{L[O|P]}{L[i|P]}$$

$$= \frac{L[IR]}{L[S(t)]} = \frac{L[SR]}{L[u(t)]} = \frac{L[RR]}{L[t, u(t)]} = \frac{L[PR]}{L[\frac{t^2}{2}, u(t)]}$$

Dynamic Response
or transient

Static Response
Steady State

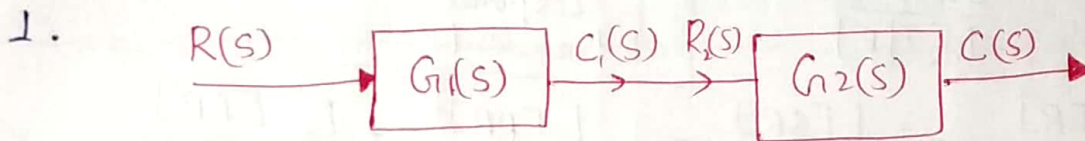
One system can't have two or more no. of transfer function. But two or more no. of systems can have same transfer function. Because transfer funⁿ doesn't depend on physical configuration.

$$\text{Transfer function} = L[IR] = SL[SR] = S^2L[RR] = S^3L[PR]$$

* BLOCK DIAGRAM Representation:

Standard topologies:

1. Series / cascade
2. Parallel / ~~feedback~~ forward.
3. feedback / canonical / closed loop.



(i) $R_2 = C_1$

(Non interactive cascading)

$$C = G_2 R_2$$

$$C_1 = R \cdot G_1$$

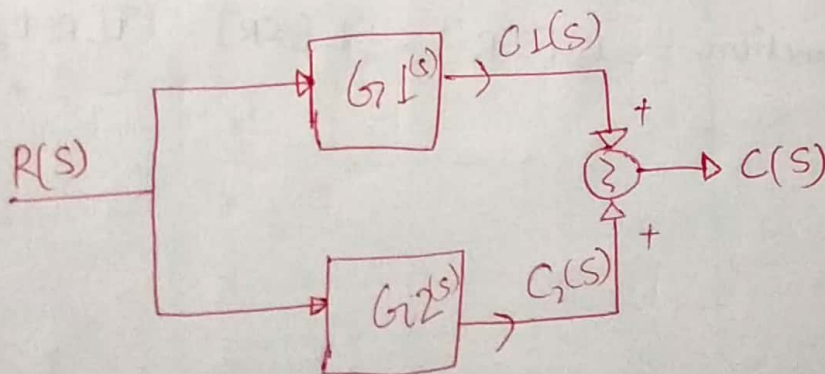
if $R_2 = C_1$

$$\boxed{\frac{C}{R} = G_1 G_2}$$

(ii) $R_2 \neq C_1$

(Interactive cascading)

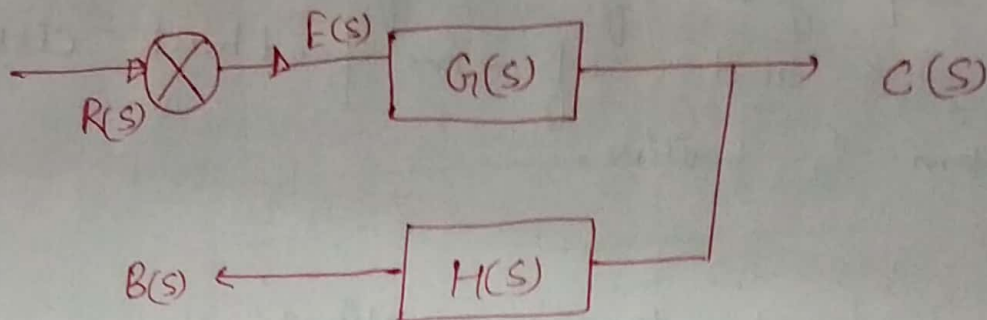
2.



$$C = C_1 + C_2 = R G_1 + R G_2$$

$$\boxed{\frac{C}{R} = G_1 + G_2}$$

3.

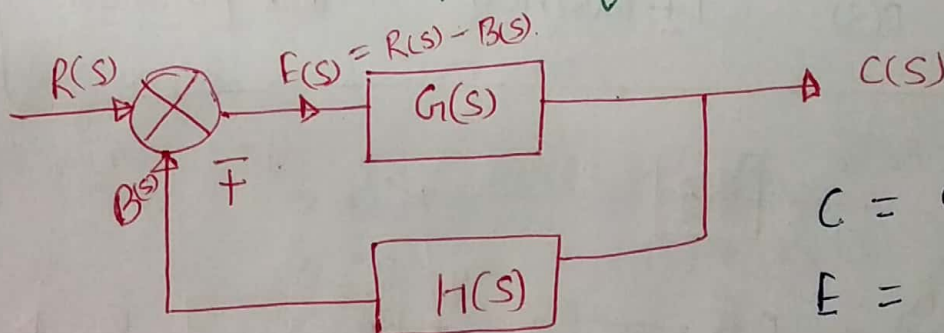


$$G(s) = \frac{C(s)}{E(s)} ; \text{ forward path T.f.}$$

$$H(s) = \frac{B(s)}{C(s)} ; \text{ feedback path T.f.}$$

$$\frac{B(s)}{R(s)} = \frac{B(s)}{E(s)} = G(s) \cdot H(s)$$

Loop transfer f. OLTf



$$C = G \cdot E$$

$$E = R - B = R - C \cdot H$$

$$E = G(R - C \cdot H)$$

$$G R = C [1 \pm G H]$$

$$\frac{C}{R} = \frac{G}{1 \pm G H}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \pm G(s) \cdot H(s)}$$

CLT.f

In closed loop transfer funⁿ

- open loop gain
- closed loop gain
- zeros poles
- open loop transfer f.

* Open loop gain of +ive feedback system is -ive and of -ive feedback system is positive. closed loop

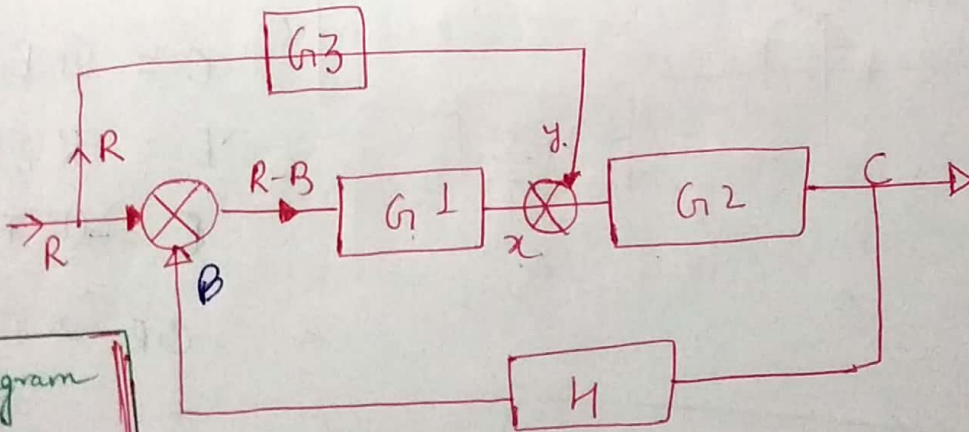
* Standard characteristics equations:-

$$q(s) = 1 \pm G(s) \cdot H(s) = 0$$

Roots of $q(s) = \text{CLP}$
(Closed loop poles)

$$\frac{C}{R} = \frac{G \cdot E}{R} = \frac{G}{1 \pm G \cdot H}$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 \pm G(s)H(s)} \quad \text{Error transfer function.}$$



If block diagram is not in any standard topology then use algebraic equation using system variables. (small diagram)

$$C = G_2 (x + y)$$

$$x = G_1 \cdot (R - B)$$

$$y = R \cdot G_3$$

$$B = C \cdot H$$

$$C = G_2 [G_1 \cdot (R - CH) + R \cdot G_3]$$

$$C = G_2 [G_1 R - G_1 G_2 H + R \cdot G_3]$$

$$C = G_1 \cdot G_2 \cdot R - G_1 \cdot G_2 \cdot G_2 H + G_2 \cdot G_3 \cdot R$$

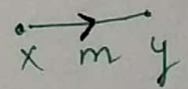
$$C [1 + G_1 \cdot G_2 H] = R [G_1 \cdot G_2 + G_2 \cdot G_3]$$

$$\boxed{\frac{C}{R} = \frac{G_2 \cdot [G_1 + G_3]}{1 + G_1 \cdot G_2 H}} \quad \underline{\text{Ans.}}$$

* Signal flow Graph :- (SFG)

↳ Both diagram and signal flow graphs are graphical representation of relation b/w variables of a system described in the form of set of linear algebraic equation.

$$y = mx$$



System is described as:

$$x(t) = x_1(t) + \beta_3 \cdot u(t)$$

$$\dot{x}_1(t) = -a_1 x_1(t) + x_2(t) + \beta_2 \cdot u(t)$$

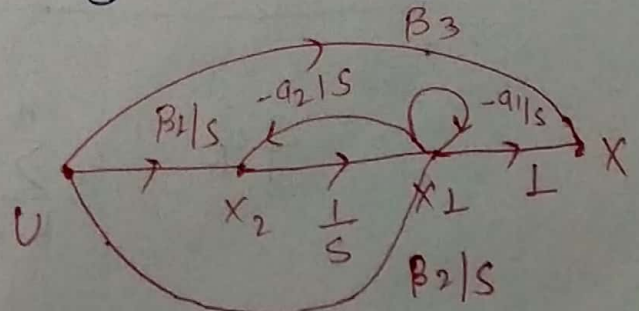
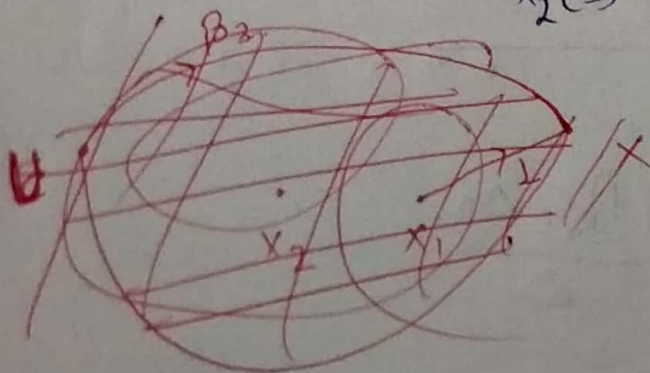
$$\dot{x}_2(t) = -a_2 x_1(t) + \beta_1 \cdot u(t)$$

L.T

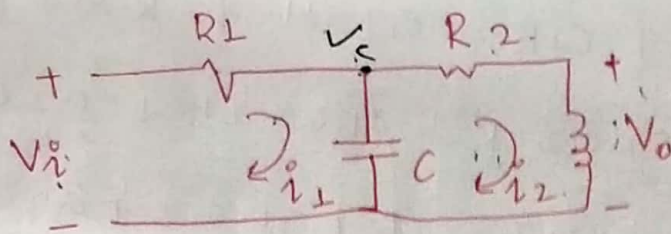
$$X(s) = X_1(s) + \beta_3 \cdot U(s)$$

$$X_1(s) = -\frac{a_1}{s} \cdot X_1(s) + \frac{1}{s} \cdot X_2(s) + \frac{\beta_2}{s} \cdot U(s)$$

$$X_2(s) = -\frac{a_2}{s} \cdot X_1(s) + \frac{\beta_1}{s} \cdot U(s)$$



Q.



$$V_i - i_1 R_1 - \frac{1}{C} \int (i_1 - i_2) dt = 0 \quad \text{--- (1)}$$

$$V_C - i_2 R_2 - \frac{1}{C} \int (i_2 - i_1) dt = 0 \quad \text{--- (2)}$$

$$- L \cdot \frac{di_2}{dt}$$

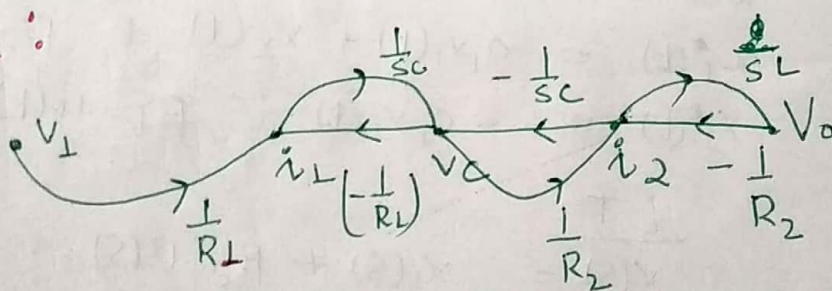
$$\Rightarrow i_1 = \frac{V_i - V_C}{R_1} \quad \text{--- (1)}$$

$$V_C = \frac{1}{sC} (i_1 - i_2) \quad \text{--- (2)}$$

$$i_2 = \frac{V_C - V_o}{R_2} \quad \text{--- (3)}$$

$$V_o = (sL) \cdot i_2 \quad \text{--- (4)}$$

SF.h :



*

Mason's formula:

$$T.F = \sum_{k=1}^n \frac{P_k \cdot \Delta_k}{\Delta}$$

$$T.F = \sum_{k=1}^n \frac{P_k \cdot \Delta_k}{\Delta}$$

$$k = 1, 2, \dots, n$$

n = No. of forward path.

P_k = Gain of k^{th} forward path.

Δ = Determinant of SFG

$$\begin{aligned} &= 1 - [\text{Sum of all individual loop gains}] \\ &+ [\text{Sop of two non-touching loop gains}] \\ &- [\text{Sop of three non-touching loop gains}] \\ &+ [\quad] \\ &- [\quad] \dots \end{aligned}$$

Δ_k = Determinant of k^{th} forward path.

$$= 1 - [\text{Sum of all individual loop gains that are non-touching to } k^{\text{th}} \text{ forward path}]$$

$$+ [\text{Sop of two non-touching loop gains that are not touching to } k^{\text{th}} \text{ forward path}]$$

$$- [\quad]$$

$$+ [\quad] \dots$$

* Node

A node in signal flow graph represents a variable of the system, no. of nodes = No. of system variables.

* Input Node:

The node that has only outgoing branches.

* Output Node:

The node that has only incoming branches.

NOTE: Mason formula is applicable only b/w input + output nodes.

Imp.

* Any variable, any no. of variables of a system can be defined as output but not as input.

* forward path:

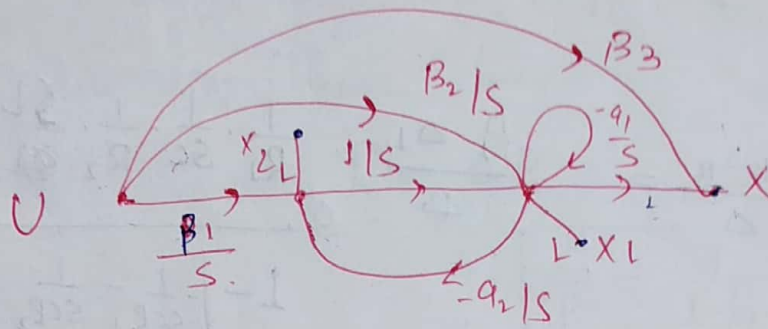
The path that connects input node with output node such that no node is repeated.

* loop:

A loop is a closed path that starts and ends at same node such that no node is repeated.

* Non-Touching ^{Loops} Node:

Two or more no. of loops are said to be non-touching if they don't have any common node.



$$\frac{X}{U} = \sum_{k=1}^3 \frac{P_k \Delta_k}{\Delta} = \frac{P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3}{\Delta}$$

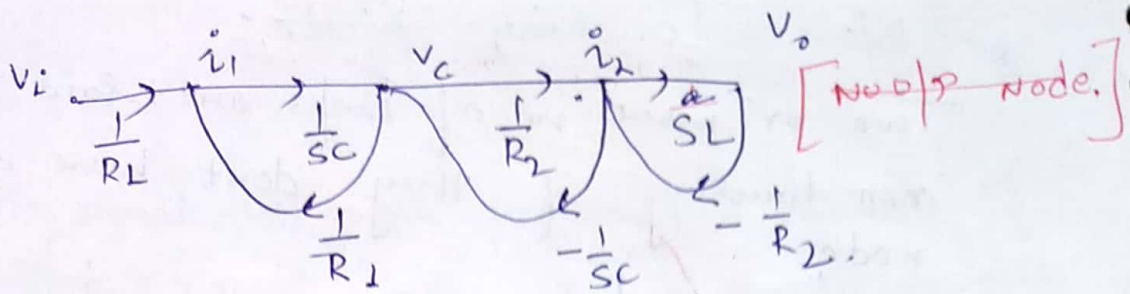
$$= \frac{\beta_1}{s^2} [1 - [0]] + \frac{\beta_2}{s} [1 - (0)] + \beta_3 [1 - (-\frac{q_2 - q_1}{s^2 s}) + [0]]$$

$$1 - [-\frac{q_1}{s} - \frac{q_2}{s^2}] + [0]$$

$$\frac{X_1}{U} = \sum_{k=1}^2 \frac{P_k \Delta_k}{\Delta} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

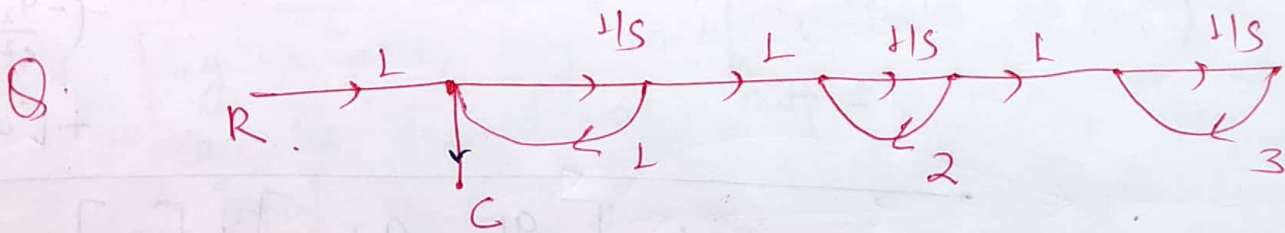
$$= \frac{\frac{\beta_2}{s} [1 - (0)] + \frac{\beta_1}{s^2} [1 - (0)]}{1 - [-\frac{q_1}{s} - \frac{q_2}{s^2}] + [0]}$$

$$\frac{X_2}{U} = \frac{\frac{\beta_1}{s} [1 - [\frac{q_1}{s}]] - \frac{q_2 \beta_2}{s^2} [1 - (0)]}{1 - [-\frac{q_1}{s} - \frac{q_2}{s^2}] + [0]}$$



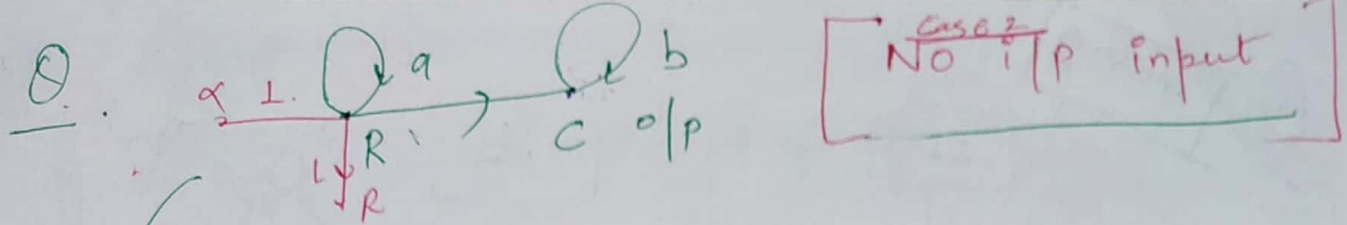
$$\frac{V_o}{V_i} = \sum_{k=1}^L \frac{P_k \Delta_k}{\Delta} = \frac{P_1 \Delta_1}{\Delta} = \frac{\frac{1}{R_1} \cdot \frac{1}{sC} \cdot \frac{1}{R_2} \cdot \frac{SL}{SL} [1 - 0]}{1 - \left[\frac{1}{sCR_1} - \frac{1}{sCR_2} - \frac{SL}{R_2} \right] + \left[-\frac{1}{sCR_1} \times -\frac{SL}{R_2} \right]}$$

$$\frac{i_2}{V_i} = \frac{\frac{1}{sCR_1} [1 - 0]}{\Delta} \quad \text{Ans}$$



$$\frac{C}{R} = \sum_{k=1}^L \frac{P_k \Delta_k}{\Delta} = \frac{1 \left[1 - \left(\frac{2}{s} + \frac{3}{s} \right) + \left(\frac{6}{s^2} \right) \right]}{1 - \left[\frac{1}{s} + \frac{2}{s} + \frac{3}{s} \right] + \left[\frac{2}{s^2} + \frac{3}{s^2} + \frac{6}{s^2} \right] - \left[\frac{6}{s^2} \right]}$$

Ans.

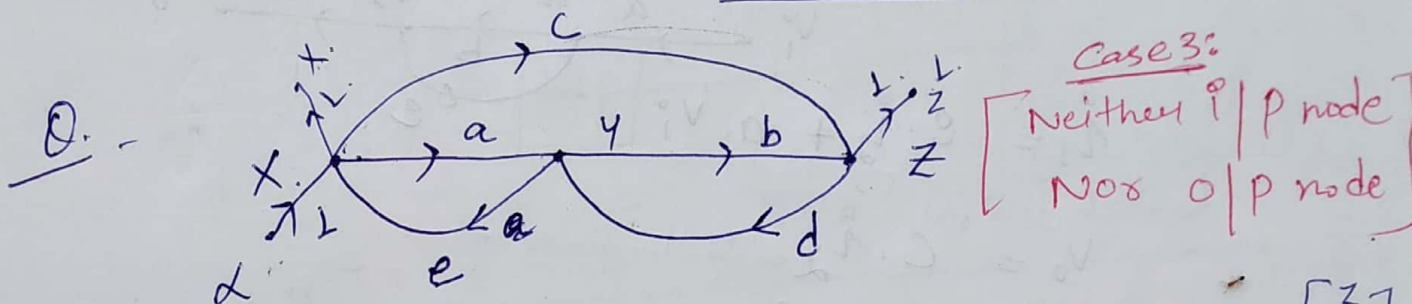


$$\frac{C}{R} = \frac{C/K}{R/K} = \frac{1(1-b)}{1(1-b)} = \frac{1}{1-b} \text{ Ans}$$

Can also be solved by algebraic Eqn:-

$$R = a \cdot R, \quad C = R + b \cdot C$$

$$\Rightarrow \boxed{\frac{C}{R} = \frac{1}{1-b}}$$



$$z = b y + c x$$

$$y = a \cdot x + d z \Rightarrow y = \frac{a}{1-bd} x$$

$$x = e \cdot y$$

$$\left[\frac{z}{x} \right]$$

$$\boxed{\frac{z}{x} = \frac{1-ae}{ed}} \text{ A}$$

$$z = b \cdot (ax) + cx$$

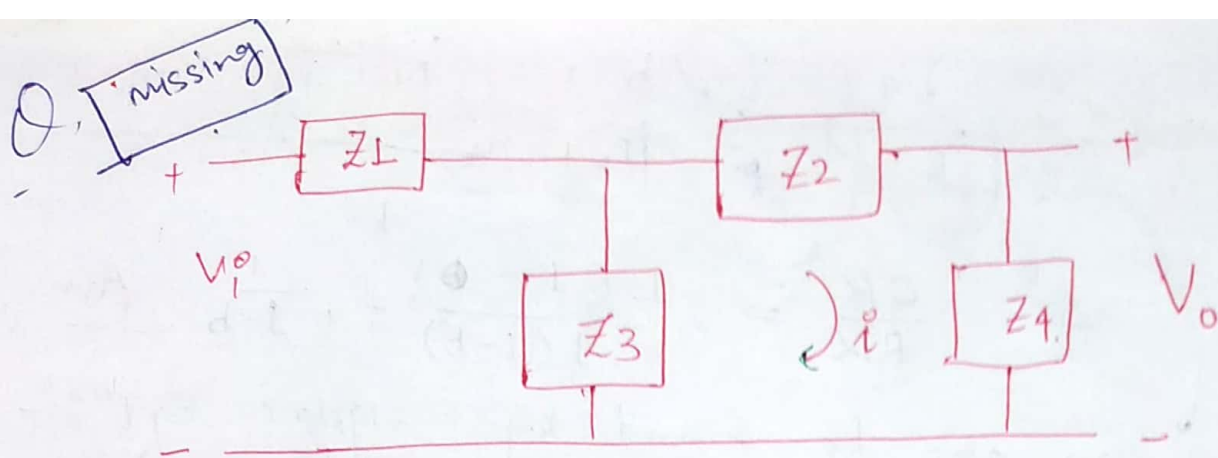
$$\frac{z}{x} = \frac{ba}{1-bd}$$

$$\boxed{\frac{z}{x} = \frac{ab+c}{1-bd}} \text{ A}$$

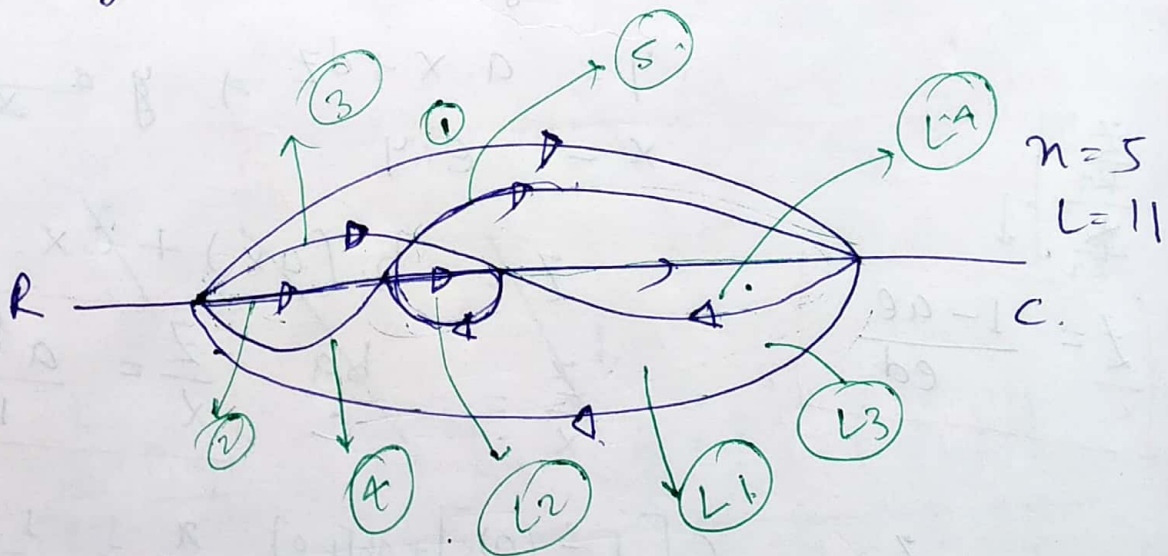
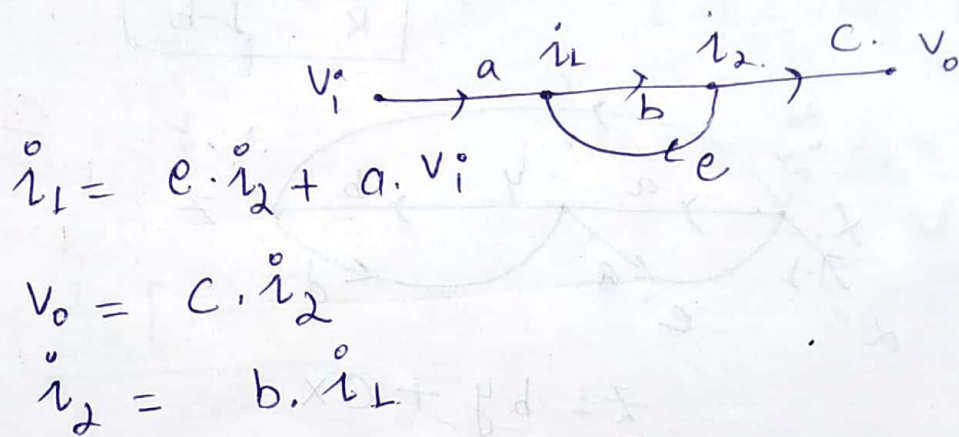
$$\frac{z}{x} = \frac{c[1-(0)] + ab[1-0]}{1 - [(ae+bd)(1-0)]}, \quad \frac{z}{x} = \frac{1(1-bd)}{1-[ae+bd]}$$

$\frac{z}{x}$ is not transfer function
hence need
not to be
unique

$$\boxed{\frac{z}{x} = \frac{ab+c}{1-bd}} \text{ Ans}$$



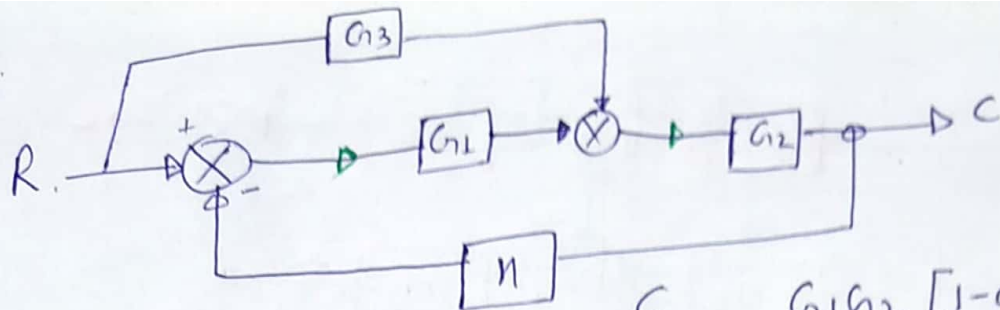
find the values of a, b, c, d .



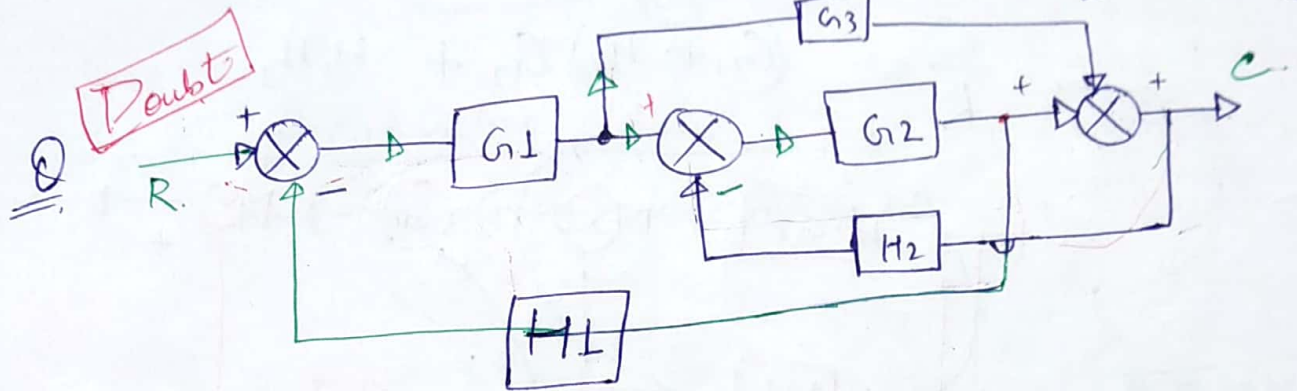
while applying Mason's formula in block diagram remember the sign of feedback while assigning the gain in loop.

NOTE:

Q.

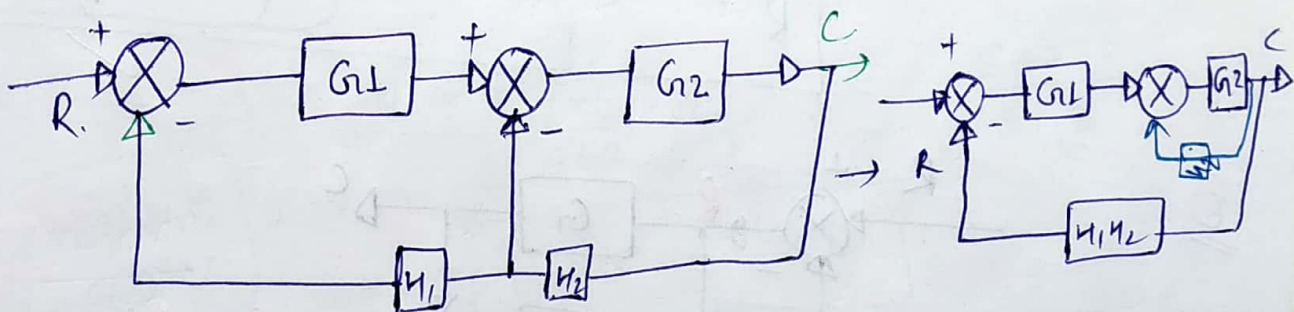


$$\frac{C}{R} = \frac{G_1 G_2 [1-0] + G_3 G_2 [1-0]}{1 - [-G_1 G_2 H]}$$



$$\frac{C}{R} = \frac{G_1 G_2 [1-0] + G_1 G_3 [1-0]}{1 - [-G_1 G_2 H_1 - G_2 H_2 + G_1 G_2 G_3 H_1 H_3] + 0}$$

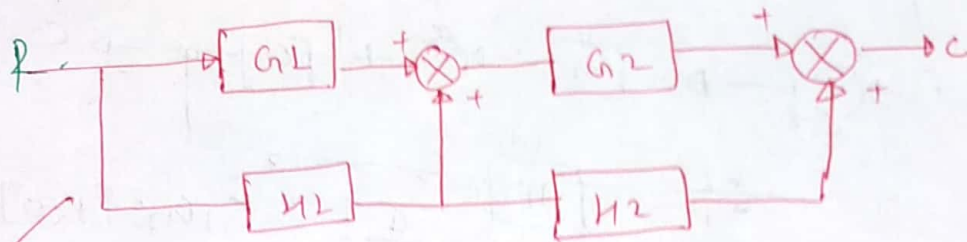
Q.



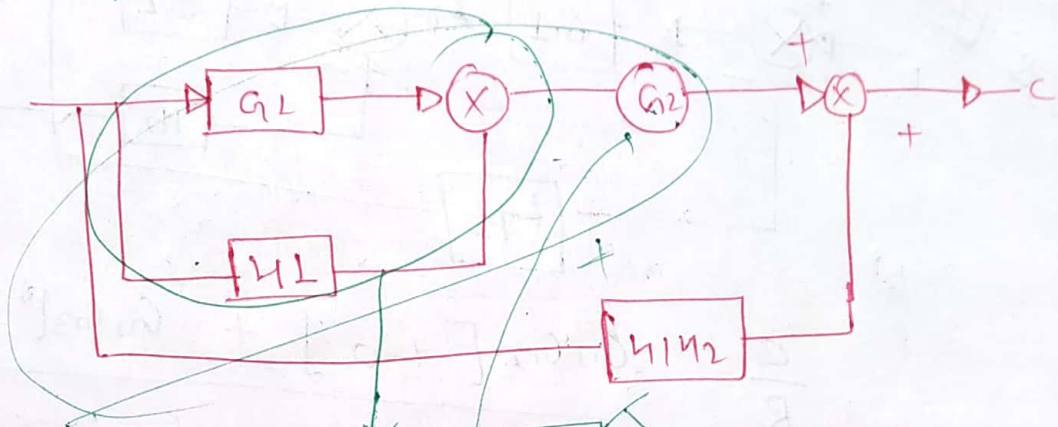
$$\frac{C}{R} = \frac{G_1 G_2 (1-0)}{1 - (-G_2 H_2 - G_1 G_2 H_1 H_2) + 0}$$

NOTE:

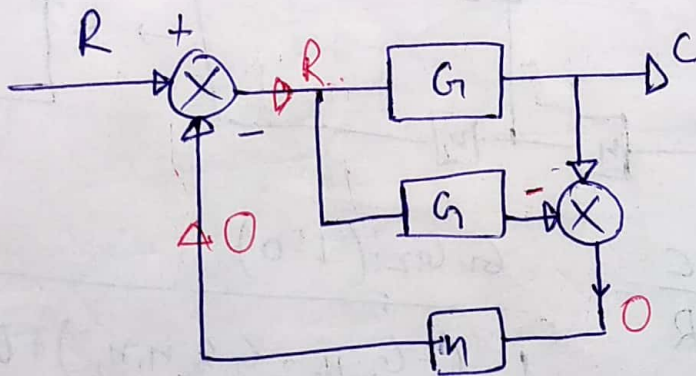
while forming T.F always consider
Sign of feedback.



$$\frac{C}{R} = (G_1 + H_1) G_2 + H_1 H_2$$



$$\frac{C}{R} = (G_1 + H_1) G_2 + H_1 H_2$$

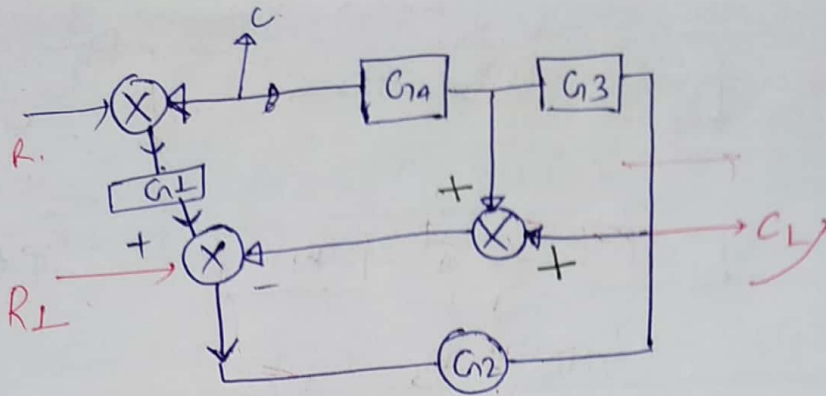


$$C = R G_2$$

$$\frac{C}{R} = G_2$$

$$\frac{C}{R} = \frac{G_1 (1 - 0)}{1 - (0 - G_1 H_1 + G_1 H_1) + 0} = G_1$$

Q

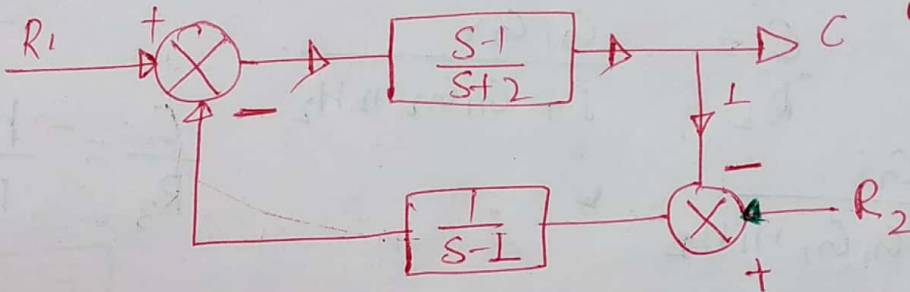


$$\frac{C}{R} = \frac{G_1 G_2 G_3 G_4}{1 - (-G_2 - G_1 G_2 G_3 G_4 - G_2 G_3) + 0}$$

$$\frac{G_2 \cdot G_3}{1 + G_2} \cdot \frac{G_4}{1 + G_2} = \frac{G_2 \cdot G_3 \cdot G_4}{1 + G_2 + G_2 G_3}$$

$$\frac{G_2 \cdot G_3 \cdot G_4 \cdot G_1}{1 + G_2 + G_2 G_3} = \frac{G_1 G_2 G_3 G_4}{1 + G_2 + G_2 G_3}$$

Q



Out to be ask
Can is Mason's
formula is
applicable
here?

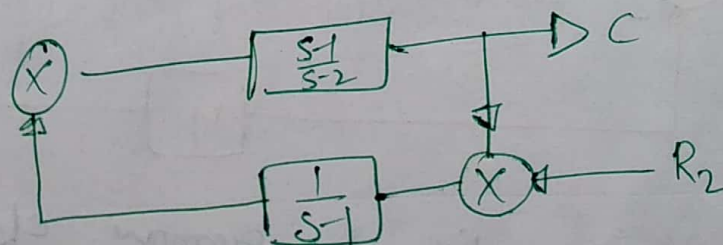
$$\Delta = 1 - \frac{s-1}{s+2} \times \frac{1}{s-1} = \frac{s+2-1}{s+2} = \frac{s+1}{s+2} = 0$$

$$q(s) = 1 + G(s)H(s)$$

$$q(s) = 1 - G(s)H(s)$$

$$q(s) = s+1 = 0$$

When R_1 is off

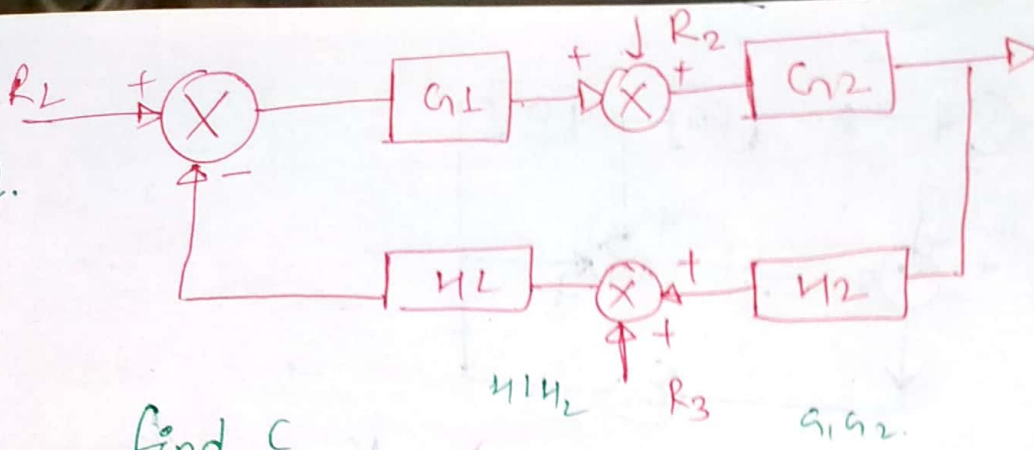


$$G(s) = -\left(\frac{1}{s-1}\right) \times \left(\frac{s-1}{s+2}\right)$$

$$H(s) = 1$$

$$q(s) = 1 + \left(-\left(\frac{1}{s-1}\right) \times \frac{(s-1)}{(s+2)}\right) = 0 = s+1$$

Q.



find C.

$$\frac{C}{R_L} = \frac{G_1 G_2}{1 + G_1 G_2 H_1 H_2}$$

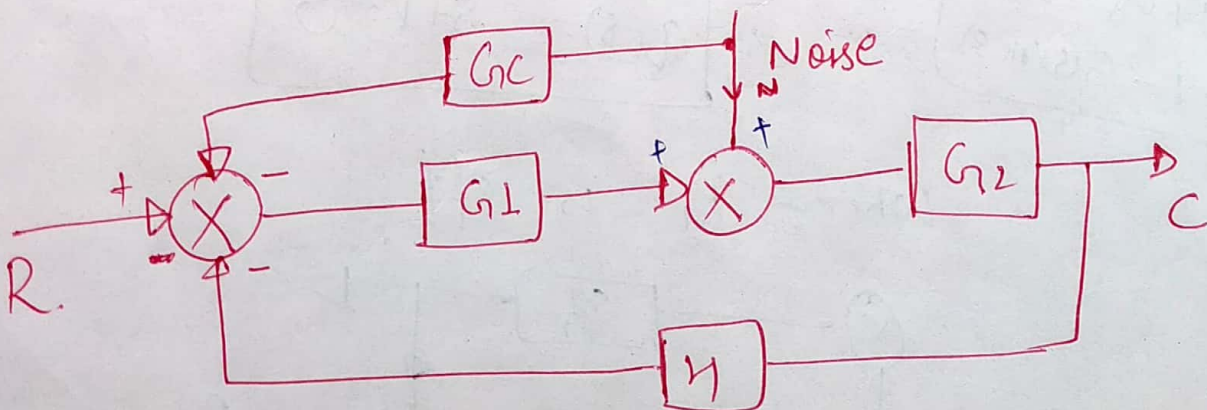
$$\frac{C}{R_L} = \frac{G_1 G_2}{1 + G_1 G_2 H_1 H_2}$$

$$\frac{C}{R_L} = \frac{G_2}{1 + G_1 G_2 H_1 H_2}$$

$$\frac{C}{R_3} = - \frac{H_1 G_1 G_2}{1 + G_1 G_2 H_1 H_2}$$

Add all

$$\Rightarrow C = \frac{G_1 G_2 R_L + G_2 R_2 - H_2 G_1 G_2 R_3}{1 + G_1 G_2 H_1 H_2}$$



Q. obtain the condition to remove effect of noise from the output of a system.

$$\frac{C}{R} = \frac{G_1 G_2}{1 + G_1 G_2 H}$$

Doubt

$$\frac{C}{N} = \frac{G_2 - G_c G_1 G_2}{1 + G_1 G_2 H}$$

$$1 - G_c G_1 G_2 = 0$$

$$G_c = \frac{1}{G_1}$$

$$C = \frac{G_1 G_2 R + G_2 [1 - G_c G_1] N}{1 + G_1 G_2 H}$$

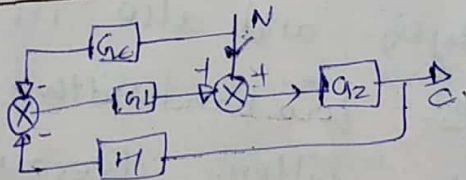
Ch-1 WB/

1. * A. (a) 2. (b) * 4. (b) 6. a
 B. (b) * 3. (b) 5. c 7. d
 8. (b) 9. (c) 10. (b) 11. (d) 12. (d)
 13. (c) 14. (d) 15. (a) 16. (c) 17. (b)
 18. 4/9 19. 9.

Ans-7

Ans-1

Ans-8



$$\frac{C}{N} = \frac{G_2 [1 - G_c G_1] + G_c G_1 G_2}{1 - G_1 G_2 H}$$

* Stability Time Response Analysis

(only for stable system)
(T.D / F.D)

T.D { $\delta(t)$ → shock
 $u(t)$ → position
 $t \cdot u(t)$ → velocity
 $\frac{t^2}{2} \cdot u(t)$ → Acceleration

F.D { $\sin \omega t$ → Oscillation.
 i/p.

System
 Order of System.
 1st, 2nd, 3rd, 4th
 exact
 prototype system.
 Approximates
 cascade of 1st + 2nd order
 (90%)

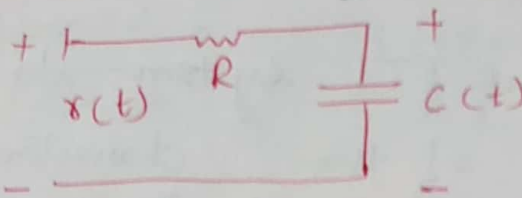
* Time Response Analysis of 1st Order System:-

Noise & B.W ~~low~~

Control System in Analysis and also in practice is preferred to be low bandwidth system. Like low pass filter. Because if the bandwidth is high then the effect of noise is also high. Due to which control system fails to track the command.



$$\frac{C(s)}{R(s)} = \frac{1}{1 + sRC}$$



$$\frac{C(s)}{R(s)} = \frac{1}{1 + sT}$$

$RC \rightarrow \text{time const.}$

Standard 1st order T. F.

* Impulse Response: $\delta(t) \longleftrightarrow 1$

$$C(s) = \frac{1}{1 + sT} \times 1$$

Amplitude

$$C(t) = \left(\frac{1}{T} \right) \cdot e^{-t/T} \cdot u(t)$$

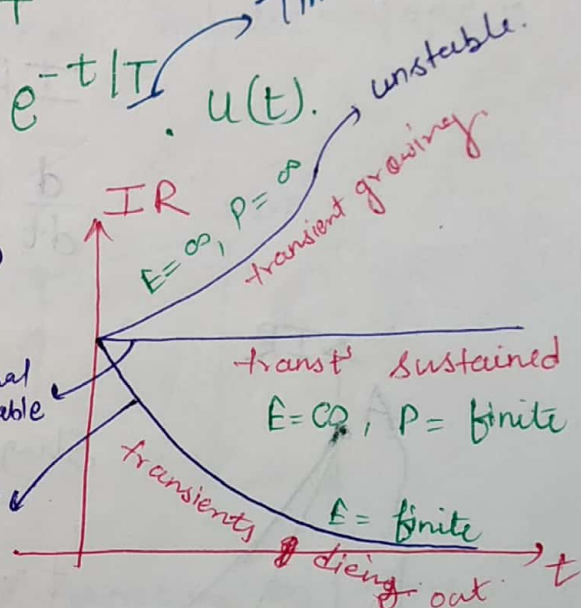
Time period.

$$\text{if } \int_0^{\infty} |IR| \cdot dt < \infty$$

then stable.

marginal stable

stable



Impulse Response of any system describes transient behaviour of the system. In a stable system transience must die out. So the system can reach to steady state. Hence if the impulse response of the system is absolutely integrable i.e.

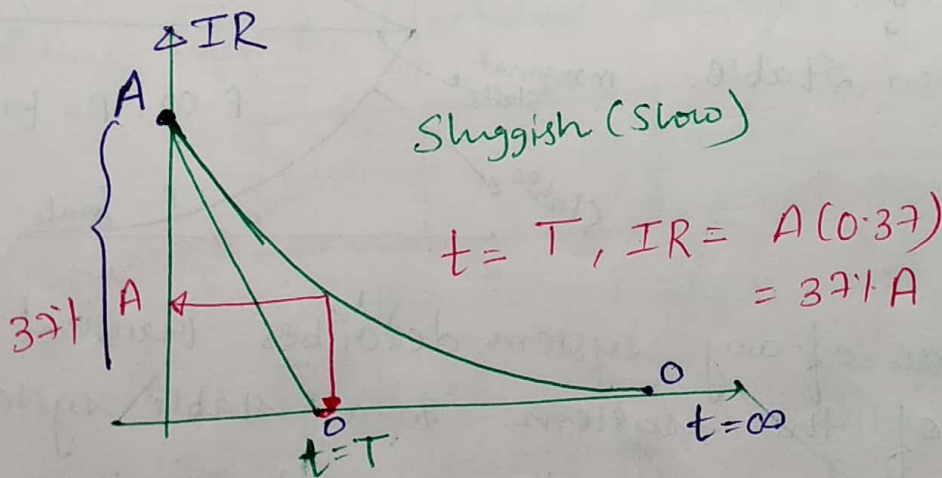
$$\int_0^{\infty} |IR| \cdot dt < \infty \rightarrow \text{Stable}$$

for a causal LTI system to be stable all the roots of the characteristic equation i.e. all the closed loop poles must be located in left side of S-plane.

Temperature Control System, level control system & speed control system are the examples of first order system.

$$IR = A \cdot e^{-t/T}$$

$$\frac{d}{dt}(IR) = A \cdot e^{-t/T} \left(-\frac{1}{T}\right) = -\frac{A}{T} \quad \text{at } t=0$$



Ideally, time constant is the amount of time required for 100% transience to die out. But practically it will be 63%.

Step Response :

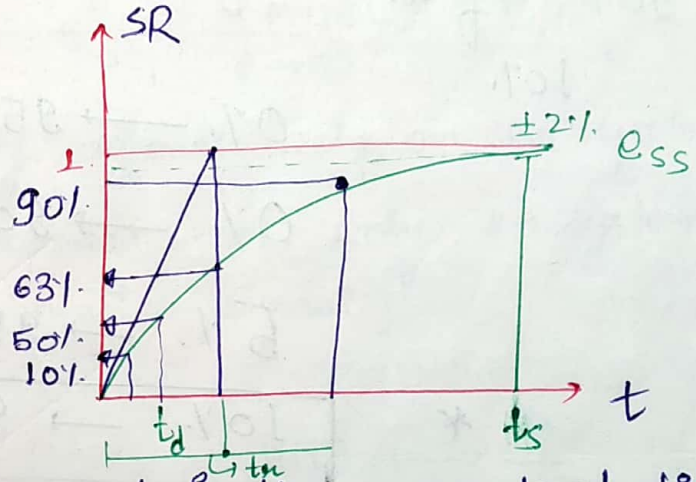
$$V(t) = \frac{1}{s}$$

$$C(s) = \frac{1}{1+sT} \times \frac{1}{s}$$

$$C(t) = \left[\underbrace{1}_{\text{Steady State}} - \underbrace{e^{-t/T}}_{\text{transients}} \right] \cdot V(t)$$

$$C(0) = 0, \quad C(\infty) = 1$$

Though stability of a system can be determined either from an impulse or step response, impulse response is preferred.



Practically, time constant is the amount of time required for a system to attain 63% of steady state.

Time required for step response to reach 50% of steady state is called delay time.

Delay time $t_d = 0.7T$ $\rightarrow$ system parameter
Rise time $t_r = 2.2T$ Response parameter.

Setting time $t_s = 4T$

Steady state error $e_{ss} = 0$

$$\begin{aligned}
 t_d &= 0.7T & : t_d \propto \text{sluggier} \\
 t_r &= 2.2T & : t_r \propto \frac{1}{\text{speed}} \\
 t_s &= 4T & : t_s \propto \frac{1}{\text{stability}} \\
 e_{ss} &= 0 & : e_{ss} \propto \frac{1}{\text{Accuracy}}
 \end{aligned}$$

* : Rise Time :

Time Taken by Step response to reach to 90% of its final value starting from 10%.

$$0\% \rightarrow 95\% : 3T$$

$$0\% \rightarrow 90\% : 2.3T$$

$$5\% \rightarrow 95\% : 2.25T$$

*

$$10\% \rightarrow 90\% : 2.2T$$

* Settling time: time required for step response to reach and to stay within a specified tolerance band.

<u>t_s</u>	<u>tolerance</u>
$1T \rightarrow 63\%$	37%
$2T \rightarrow 86\%$	14%
$3T \rightarrow 95\%$	5%
$4T \rightarrow 98\%$	2%
$5T \rightarrow 99.3\%$	0.7%
$\infty T \rightarrow 100\%$	0%

* Steady State error:

It is the difference b/w ~~desired~~ input value and steady state value of o/p.

$$e(t) = R(t) - C(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

$$e(t) = U(t) - [1 - e^{-t/T}] \cdot U(t).$$

$$e(t) = e^{-t/T} \cdot U(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = 0$$

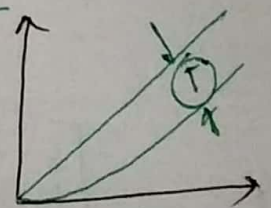
Speed and stability depends only on the system but accuracy depends not only on system but also on input signal.

* Ramp Response:

$$t \cdot U(t) \longleftrightarrow \frac{1}{s^2}$$

$$C(s) = \frac{1}{1+ST} \times \frac{1}{s^2}$$

$$C(t) = \left[\underbrace{t - T}_{\text{Steady state}} + \underbrace{T \cdot e^{-t/T}}_{\text{Transient}} \right] \cdot U(t)$$

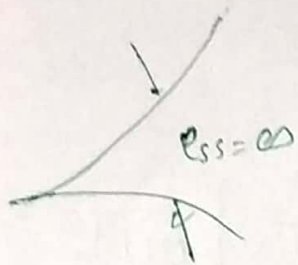


$$e(t) = R(t) - C(t) = T [1 - e^{-t/T}] \cdot U(t).$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = T \neq 0$$

* Parabolic Response: $\frac{t^2}{2} \cdot u(t) \longleftrightarrow \frac{1}{s^3}$

$$C(s) = \frac{1}{1+sT} \times \frac{1}{s^3}$$



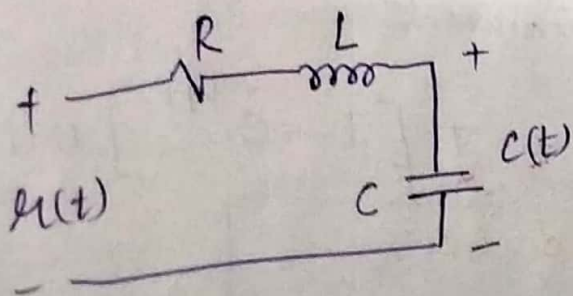
$$C(t) = \left\{ \frac{t^2}{2} - T \left[t - T(1 - e^{-t/T}) \right] \right\} \cdot u(t)$$

$$e(t) = u(t) - c(t) = T \left[t - T(1 - e^{-t/T}) \right] \cdot u(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \infty$$

Standard first order system is ~~capable~~ capable of tracking position but not velocity and acceleration.

* Time Response Analysis 2nd Order:



$$\frac{C(s)}{R(s)} = \frac{1}{s^2 LC + sRC + 1}$$

* Standard 2nd Order System:

$$\boxed{\text{T.f } \frac{C(s)}{R(s)} = \frac{\omega_n^2 [x]}{1 \cdot s^2 + 2\{\zeta \omega_n s + \omega_n^2\}}}$$

ω_n = Natural frequency / undamped frequency of oscillations.

ζ = Damping ratio

Except first Order System, all other systems exhibits oscillations due to which they are not stable.

In Order to stabilize the system, these oscillations are killed by means of mechanism called damping. Hence control system in practice is always damped.

$$\checkmark \text{ T.f } = \frac{5}{1 \cdot s^2 + 2 \cdot s + 5}, \quad \text{T.f} = \frac{5}{2s^2 + 2s + 5} = \frac{5/2}{s^2 + s + 5/2}$$

$\omega_n = \sqrt{5}$ $\omega_n = \sqrt{\frac{5}{2}}$

$$\times \text{ T.f } = \frac{1}{s^2 + 2s + 5}, \quad \text{T.f} = \frac{5(s+1)}{s^2 + 2s + 5} \times$$

coefficient of s^2 should be unity, & Value of Numerator should always equal to constt value in denominator.

* Standard 2nd Order System:

$$\text{T.f } \frac{C(s)}{R(s)} = \frac{\omega_n^2 [\times]}{1 \cdot s^2 + 2\{\omega_n s + \omega_n^2\}}$$

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coefficient of s^2 should be unity, & Value of Numerator should always equal to constt value in denominator.

$$\frac{C}{R} = \frac{\text{gain}}{s^2 + s \cdot \frac{R}{L} + \frac{1}{LC}} \rightarrow \text{frequency}^2$$

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$2\zeta\omega_n = \frac{R}{L}$$

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

* Damping Method:

$$q(s) = s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

$$s_{1,2} = f[\omega_n, \zeta]$$

$0 < \zeta < \infty$; positive Damping.

$-\infty < \zeta < 0$; -ive Damping.

Practically, control system is positively damped because
-ive damped system is unstable.

①. $\zeta = 0$ [undamped system] ③ $\zeta = 1$
 $s_{1,2} = 0 \pm j\omega_n$; imaginary k.b (marginally stable)
 $s_{1,2} = -\omega_n$

②. $0 < \zeta < 1$ ④. $1 < \zeta < \infty$
 $s_{1,2} = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}$
 complex conjugate in left side of plane.

②.

$$0 < \rho < 1 \quad \text{damping factor } (\alpha)$$

$$s_{1,2} = -\boxed{\rho \omega_n} \pm j \boxed{\omega_n \sqrt{1-\rho^2}} \quad \omega_d$$

under-damped system.

Complex conjugate in left side of s-plane.

$$\boxed{\alpha = \rho \omega_n ; \text{ Damping factor.}}$$

$$\boxed{\omega_d = \omega_n \sqrt{1-\rho^2}} ; \text{ Damped (or) underdamped freq. of oscillation.}$$

$$\alpha = \rho \omega_n$$

$$\boxed{\rho = \frac{\alpha}{1 \cdot \omega_n}} = \frac{\text{Actual damping factor}}{\text{critical damping factor}}$$

③.

$$\rho = 1 : \text{ [perfectly or critically damped system]}$$

$$s_{1,2} = -\omega_n, -\omega_n ; \text{ real, equal, -ive.}$$

④.

$$1 < \rho < \infty : \text{ [Overdamped system]}$$

$$s_{1,2} = -\rho \omega_n \pm \omega_n \sqrt{\rho^2 - 1} \quad \text{zeta}$$

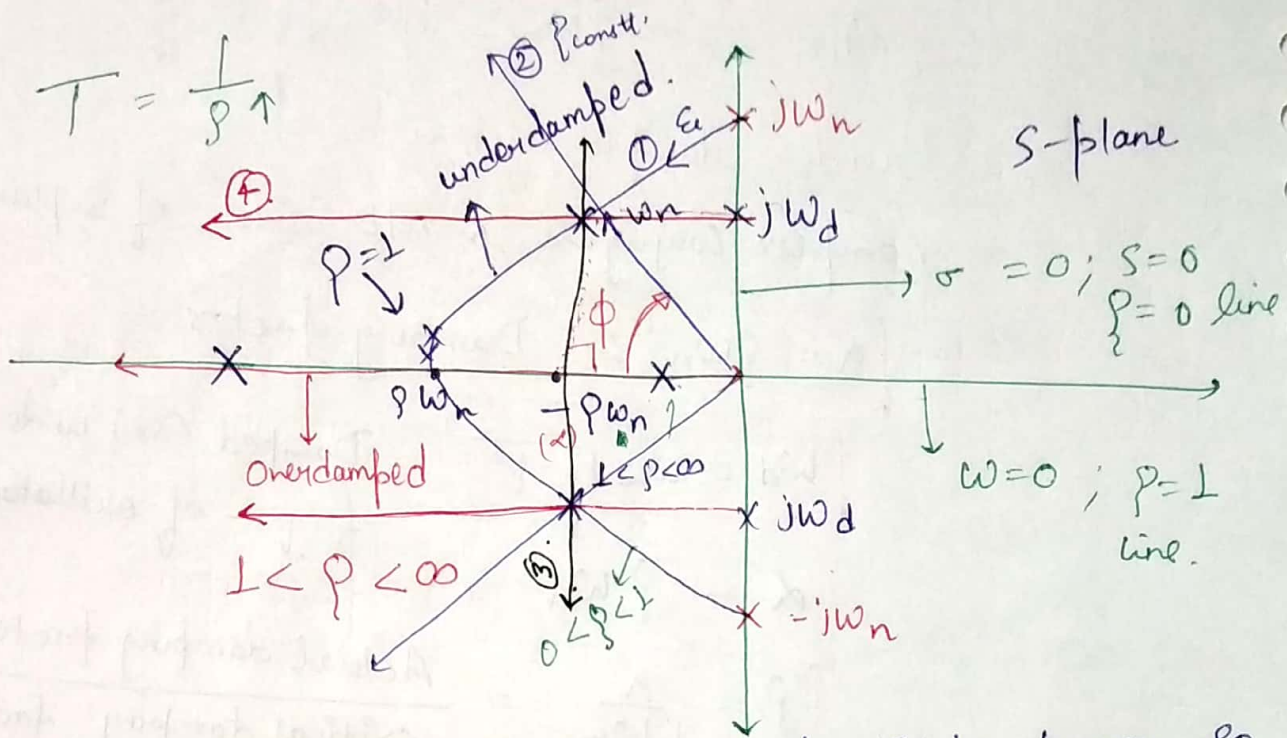
Roots are Real, unequal, -ive.

Sluggish.

$\therefore$ Time constt will be high $\left[T = \frac{1}{\rho} \right]$
hence speed $\downarrow$, stability $\downarrow$ new.

(How) $\rho \rightarrow$ distance of pole from origin

* Root Locus of 2nd Order System



→ In over-damped system time constant will be larger so it will be slow.

* → Undamped system is not stable.

→ Critically damped system is impractical.

→ Time constant of the over-damped system becomes huge (due to dominant root) due to which speed and stability decreases of the system.

→ Overdamped system is practically 2nd order but functionally first order.

→ Time constant of the under-damped system decreases due to which speed and stability increases hence control system in practice is implemented to be underdamped system.

$$T_{\text{under-damped}} = \frac{1}{\rho \omega_n} \text{ , if pole is real, } T = \frac{1}{\text{real part}}$$

$$\text{if pole is complex, } T = \frac{1}{\text{real part}}$$

$$T_{\text{over-damped}} = \frac{1}{\rho \omega_n - \omega_n \rho \pm 1}$$

$$\omega_n \cos \phi = \alpha$$

$$\Rightarrow \omega_n \cos \phi = \rho \omega_n$$

$$\omega_n \sin \phi = \omega_d$$

$$\boxed{\rho = \cos \phi}$$

$$\sqrt{\alpha^2 + \omega_d^2} = \omega_n$$

$$\omega_n \sin \phi = \omega_n \sqrt{1 - \rho^2}$$

$$\boxed{\sin \phi = \sqrt{1 - \rho^2}}$$

①. Constant ω_n ; $0 < \rho < 1$

②. Constant ρ ; $0 < \omega_n < \infty$

③. Constant ω_d ; $0 < \alpha < \infty$

④. Constant α ; $0 < \omega_d < \infty$

By default
2nd Order Syst^m
is under-damped

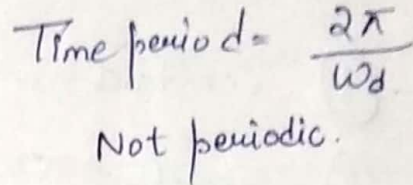
* Impulse Response :

$$C(s) = \frac{\omega_n^2}{s^2 + 2\rho\omega_n s + \omega_n^2} \times 1$$

$$C(t) = \underbrace{\frac{\omega_n}{\sqrt{1 - \rho^2}}}_{\text{Amplitude}} e^{-\rho\omega_n t} \sin \omega_d t \cdot U(t).$$

$$\boxed{\omega_n = \sqrt{\alpha^2 + \omega_d^2}}, \quad \boxed{\rho = \frac{\alpha}{\omega_n}}$$

$$C(t) = \frac{\omega_n}{\sqrt{1 - \rho^2}} \cdot e^{-\rho\omega_n t} \sin \omega_d t \cdot U(t)$$



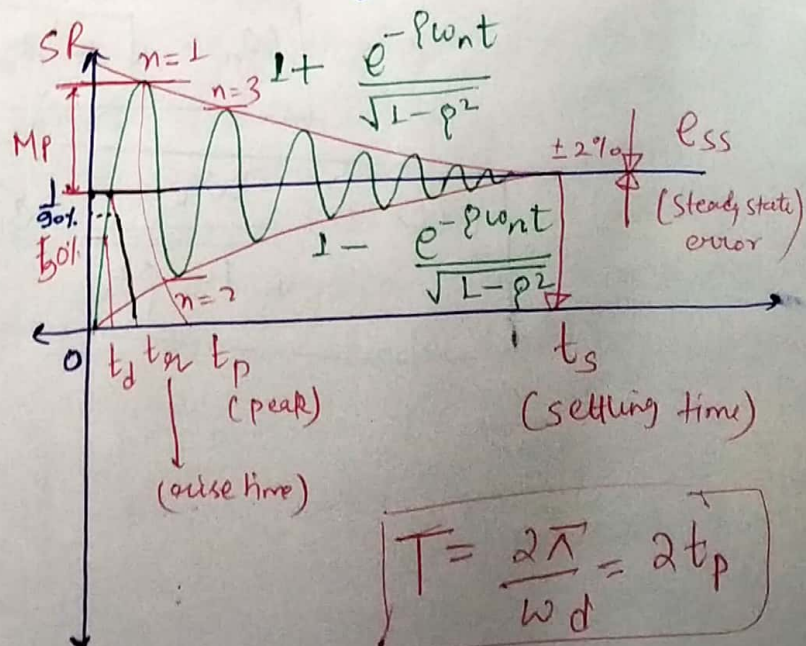
Step Response :

$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \times \frac{1}{s}$$

$$C(t) = \left[\underbrace{1}_{\substack{\uparrow \\ \text{Steady} \\ \text{State}}} - \underbrace{\frac{e^{-\rho \omega t}}{\sqrt{1-\rho^2}} \cdot \sin(\omega t + \phi)}_{\substack{\text{Steady state} \\ \uparrow \\ \text{transient state}}} \right] u(t)$$

$$C(0) = 10$$

$$C(\infty) = 1$$

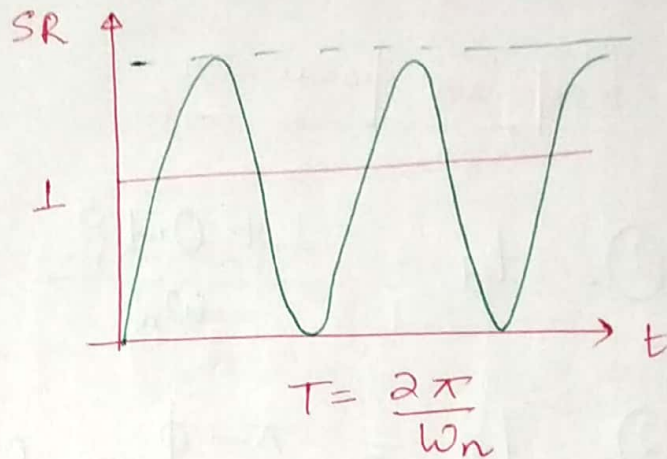


$M_p \rightarrow \text{Peak}^{\text{over}} \text{Shoot.}$

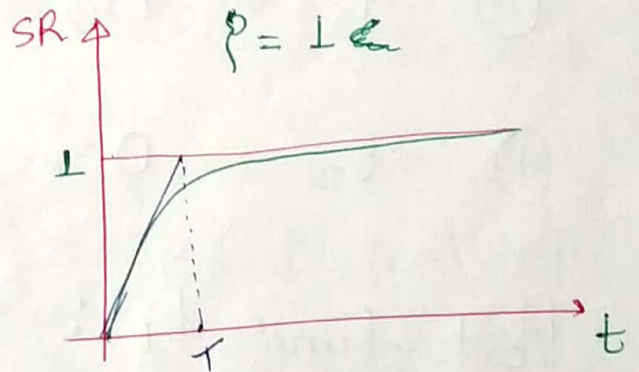
$$T = \frac{2\pi}{\omega_d} = 2t_p$$

* for Un-damped System

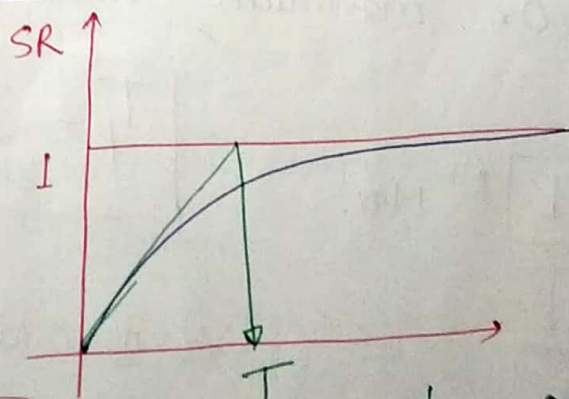
$$\zeta = 0$$



* for Critically damped System:



* for Over-damped System



$$T_{crit} < T_{over}$$

$$\begin{aligned} t_d &= 0.7 T \\ t_{cr} &= 2.2 T \\ t_s &= 4 T \end{aligned}$$

$\omega_n, \zeta \rightarrow$ System parameter.

Response parameter

$$(1). t_d \approx \frac{1 + 0.7\zeta}{\omega_n}$$

(0% to 100%) (2). $t_r = \frac{\pi - \phi}{\omega_d} \approx \frac{0.8 + 2.5\zeta}{\omega_n}$

$$(3). t_s = 4\tau = 4 \times \frac{1}{\zeta\omega_n} = \frac{4}{\zeta\omega_n}$$

$$(4). e_{ss} = 0$$

(5). Peak time t_p :

Time take by step ^{response} to reach to peak value
or maximum value.

$$T_p = \frac{n\pi}{\omega_d}$$

$n = 2K - 1$ k^{th} Max

$n = 2K$
 k^{th} Min

$\nabla n = \text{odd} \Rightarrow$ Maxima (or) Overshoot
 $\nabla n = \text{even} \Rightarrow$ Minima or undershoot

[for n^{th} Maxima, n^{th} odd should
be taken & for minima n^{th} even]

$$\text{No. of cycles} = \frac{t_c}{\text{period}} = \frac{\frac{4}{\rho \omega_n}}{\frac{2\pi}{\omega_n \sqrt{1-\rho^2}}} = \frac{2}{\pi} \frac{\sqrt{1-\rho^2}}{\rho}$$

$$\boxed{\text{No. of cycles} = \frac{2}{\pi} \frac{\sqrt{1-\rho^2}}{\rho}}$$

$$\text{for } \rho = 0.1, \text{ cycles} = 6.33$$

* M_p (peak Overshoot): is the difference between peak value of the response and final value

$$\begin{aligned} M_p &= C(t_p) - C(\infty) \\ &= C(t_p) - 1; \text{ unit step} \\ &= A[C(t_p) - 1]; \text{ step} \end{aligned}$$

$$\% M_p = \frac{A[C(t_p) - C(\infty)]}{A[C(\infty)]} \times 100$$

M_p changes proportionality with i/p amplitude but
 $\% M_p$ doesn't change.

$$\begin{aligned} M_p &= e^{-\left(\frac{n\pi\rho}{\sqrt{1-\rho^2}}\right)}; \text{ unit step} \\ &= A \cdot e^{-\left(\frac{n\pi\rho}{\sqrt{1-\rho^2}}\right)}; \text{ step} \end{aligned}$$

$$\% M_p = e^{-\frac{n\pi\rho}{\sqrt{1-\rho^2}}} \times 100$$

from M_p we can calculate ξ
and then calculate other parameter

$$t_p = \frac{\pi}{\omega_d}$$

$$0 < \xi < 1 \Rightarrow 1 > m_p > 0$$

$$m_p \propto \frac{1}{\text{Stability}}$$

$$t_s \propto \frac{1}{\text{Stability}}$$

t_s is more preferable.

$\therefore t_s$ do exist for overdamped, critically damped
+ 1st order system.

* Non-Standard 2nd Order System

form 1: $T.F = \left(\frac{1}{5}\right) \times \frac{5}{s^2 + 2s + 5}$, (t_d, t_r, t_p, t_s)
 $\downarrow$
 Same as Standard

Correction factor

but $M_p = A \times B \times e^{-n\pi\xi/\sqrt{1-\xi^2}}$

% M_p is always same for ~~any~~
Standard & non-Standard system.

form 2:

$$T.F : \frac{5(s+1)}{s^2 + 2s + 5} \times \frac{1}{s}$$

t_s is same

$$c(t) =$$

Derive whole expression. for

t_d, t_s, t_p, m_p

* Steady state error :

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

$$e(t) = \underset{\text{I/P}}{r(t)} - \underset{\text{O/P}}{c(t)}$$

→ inverse Laplace.
difficult to find.

$$\frac{E}{R} = \frac{1}{1+G \cdot H}$$

$$\lim_{s \rightarrow 0} s \cdot E(s)$$

$$\lim_{s \rightarrow 0} \frac{s \cdot R(s)}{1+G \cdot H}$$

I/P
OLT F

only for e_{ss}

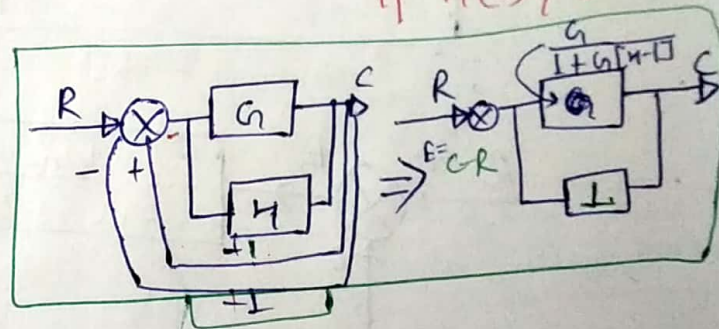
$$G(s) \text{ if } H(s)=1$$

$$\frac{G(s)}{1+G(s)[H(s)-1]}$$

if $H(s) \neq 1$

$$OLT \cdot F = G(s) \cdot H(s)$$

✓ topic of G.S
except e_{ss}



CLTF

Standard CLTF

$$* T.F = \frac{1}{1+ST} = \frac{G}{1+G}$$

$$1+G = G[1+ST]$$

$$G = \frac{1}{ST}$$

OR

$$\frac{1}{1+ST}$$

Substr ① from denominator

$$G(s) = \frac{1}{1+ST+1} = \frac{1}{ST}$$

T.F

$$\frac{1}{1+ST}$$

Add numerators to Denominators to get OLT.F in case of +ve feedback

Valid only for unity feedback control system.

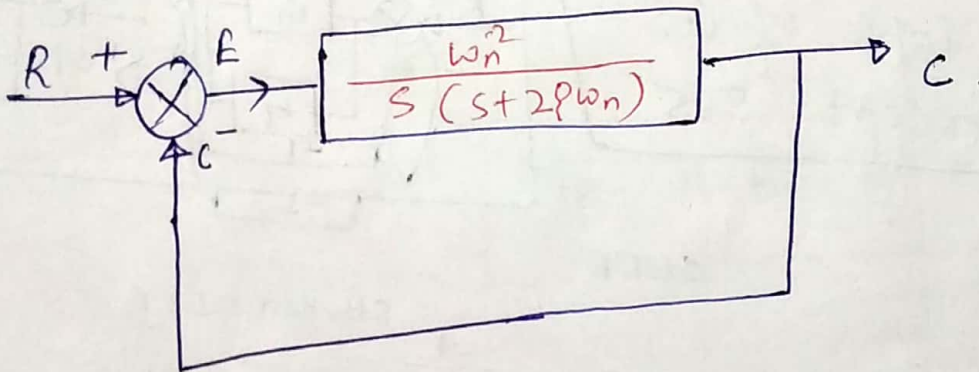
Subtract ~~Num.~~ Denominator with numerator to get OLT.F (in case of -ve feedback)

eg; $G(s) = \frac{1}{1+ST-L} = \frac{1}{ST} \text{ Ans}$

2nd order :

* T.F = $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2 - \omega_n^2}$$



* Standard 1st Order : $G(s) = \frac{1}{ST}$

(a). $R(s) = \frac{1}{s} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \times \frac{1}{1+ST} = 0$

(b). $R(s) = \frac{1}{s^2} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \times \frac{1}{1+\frac{1}{ST}} = T$

$$\textcircled{c}. R(s) = \frac{1}{s^3} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{\frac{1}{s^3}}{1 + \frac{1}{sT}} = \infty$$

Standard order T.F is of type: 1

* Standard 2nd Order: $G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$

$$\textcircled{a}. R(s) = \frac{1}{s} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \times \frac{\frac{1}{s}}{1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}} = 0$$

$$\textcircled{b}. R(s) = \frac{1}{s^2} \Rightarrow e_{ss} = \lim_{s \rightarrow 0} s \times \frac{\frac{1}{s^2}}{1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}} = \frac{2\zeta}{\omega_n}$$

$$\textcircled{c}. R(s) = \frac{1}{s^3} \Rightarrow e = \lim_{s \rightarrow 0} s \times \frac{\frac{1}{s^3}}{1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}} = \infty$$

* Time Response Parameters:

t_d

t_{ri}

t_p

t_s

m_p

transient
response
parameter

e_{ss}

Steady state
parameter

$$0 \leq \text{type} \leq \text{order}$$

$$1 \leq \text{order} \leq \text{Any}$$

Stability is determined only by
"closed loop poles."

- (i) Steady State error depends on Input signal and type of a system but not on order.
- (ii) Order of system is defined as total no. of poles. And it is applicable for both open loop & close loop.

(iii) Type of a system is defined as no. of poles at origin and it is applicable only for open loop system.

(iv) Order of a system is used to describe transient characteristics and type of a system is used to describe steady state characteristics.

Type = 0:

$$G(s) = \frac{K(s+Z_1) \dots}{s^0(s+P_1) \dots}$$

Steady state behaviour can't be studied by impulse response. Hence e_{ss} can't be calculated from IR.

(a). $R(s) = \frac{1}{s}$

$$e_{ss} = \lim_{s \rightarrow 0} s \times \frac{1}{s} \frac{1}{1 + \frac{K(s+Z_1) \dots}{(s+P_1) \dots}} = \frac{1}{1 + \frac{K Z_1 Z_2 \dots}{P_1 P_2 \dots}}$$

$$K_p = \lim_{s \rightarrow 0} G(s)$$

$$e_{ss} = \frac{1}{1 + K_p}$$

K_p - positional error const.

(b).

$$R(s) = \frac{1}{s^2}$$

K_a, K_p, K_v are also called open loop DC gain & Zero frequency gain

$$e_{ss} = \lim_{s \rightarrow 0} s \times \frac{1}{s^2} = \frac{\infty}{1+K_p} = \infty$$

(c).

$$R(s) = \frac{1}{s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s \times \frac{1}{s^3} = \frac{\infty}{1+K_p} = \infty$$

K_p, K_a, K_v is Open loop dc-gain

K_v is velocity error constt

e_{ss}	$\frac{A}{s}$	$\frac{A}{s^2}$	$\frac{A}{s^3}$
Type=0	$\frac{A}{1+K_p}$	∞	∞
=1	0	A/K_v	∞
=2	0	0	$\frac{A}{K_a}$
≥ 3	0	0	0

K_a is acceleratⁿ error constt

Residue

Steady state error or Static error

OLDC gain or Zero frequency gain

$$K_p = \lim_{s \rightarrow 0} \frac{C(s)}{1+C(s)G(s)} ; \text{positional error constt}$$

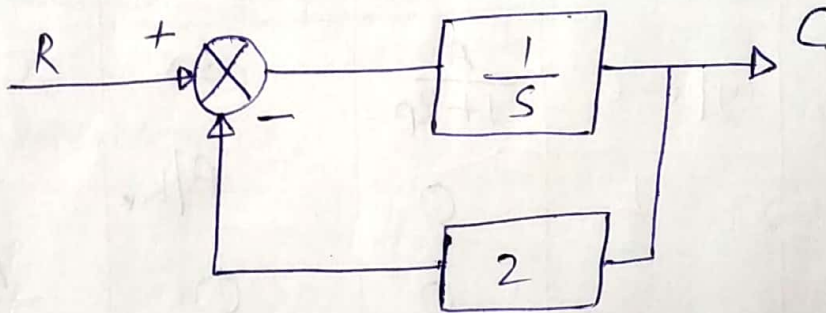
$$K_v = \lim_{s \rightarrow 0} s \frac{C(s)}{1+C(s)G(s)} ; \text{velocity " "}$$

$$K_a = \lim_{s \rightarrow 0} s^2 \frac{C(s)}{1+C(s)G(s)} ; \text{Acceleratⁿ " "}$$

put $S=0$ in T.F of system
to obtain DC gain
for open loop in OLTF & closed loop CLTF

* if the type of the system increased then
Steady state error decreases but stability
also decreases.

* Non Unity feedback Steady state error:-



$$G(s) = \frac{1}{s}$$

$$G(s) \cdot H(s) = \frac{2}{s}$$

$$1 + G(s) \cdot [H(s) - 1] = \frac{1}{s+1}$$

type = 0

$$R(s) = \frac{1}{s} \Rightarrow e_{ss} = \frac{1}{1+K_p}$$

$$K_p = \lim_{s \rightarrow 0} \frac{1}{s+1} = 1$$

$$e_{ss} = \frac{1}{2} = 0.5$$

$$\frac{C}{R} = \frac{1/s}{1 + \frac{1}{s}(2)} = \frac{1}{s+2}$$

$$C(s) = \frac{1}{s+2} \times \frac{1}{s}$$

$$C(t) = 0.5 [1 - e^{-2t}] \cdot u(t)$$

$$e(t) = r(t) - c(t)$$

$$= u(t) - 0.5u(t) + 0.5 \cdot e^{-2t} u(t)$$

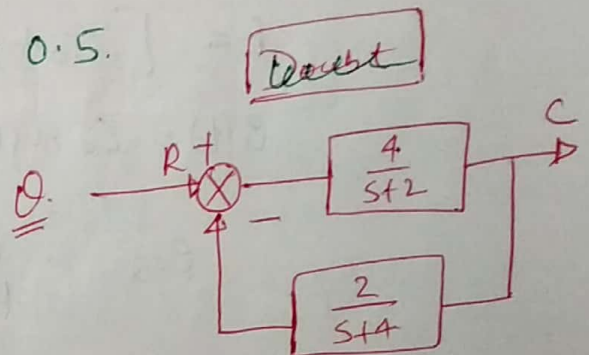
$$= 0.5 \cdot u(t) + 0.5 \cdot e^{-2t} u(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = 0.5$$

Step i/p

$$e_{ss} = A[1 - \text{CLDC gain}]$$

$$e_{ss} = 1(1 - 0.5) = 0.5$$



$$e_{ss} = \frac{1}{1 + K_p}$$

$$K_p = \lim_{s \rightarrow 0} s \cdot \frac{4}{s+2} \cdot \frac{2}{s+4}$$

$$\text{OLTF} = \frac{4}{s+2} \cdot \frac{2}{s+4}$$

$$= \frac{4 \cdot 2}{(s+2)(s+4)}$$

$$\frac{\frac{4}{s+2}}{1 + \frac{4}{(s+2)} \cdot \left[\frac{2}{s+4} - 1 \right]} = \frac{-(s+2)}{(s+4)}$$

$$\text{OLTF} = \frac{4(s+4)}{s^2 + 6s + 8 - 4s - 8}$$

$$= \frac{4(s+4)}{s(s+2)}, \text{ type} = 1$$

$$\text{CLTF} = \frac{4s+16}{s^2 + 6s + 16}$$

$$\text{CLDC gain} = 1$$

$$e_{ss} = 1(1 - 1) = 0$$

$$= 0$$

$$\frac{1}{K_v} = \frac{1}{8}$$

$$\infty$$

* Dynamic error series:

Static method of finding steady state error is applicable only for stable system, only for standard input and only at $t = \infty$. Whereas dynamic method of finding error is applicable for any system, for any input at any time.

$$\frac{E}{R} = \frac{1}{1 + OLTF}$$

$$E = [C_0 + C_1 S + C_2 S^2 + \dots] R$$

$$e(t) = C_0 r(t) + C_1 r'(t) + C_2 r''(t)$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

$$G(s) = \frac{1}{s+1}, \quad H(s) = 1, \quad \frac{E}{R} = \frac{1}{1 + \frac{1}{s+1}} = \frac{s+1}{s+2}$$

$$\frac{E}{R} = \frac{s+1}{s+2}, \quad E/R = 0.5 (0.5s + 1)^{-1} (s+1)$$

$$e(t) = 0.5 (1 - 0.5s + (0.5s)^2 - (0.5s)^3 + \dots) [(s+1)]$$

$$e(t) = \underbrace{0.5}_{C_0} s^0 - \underbrace{0.25}_{C_1} s^1 + \underbrace{0.125}_{C_2} s^2 + \dots$$

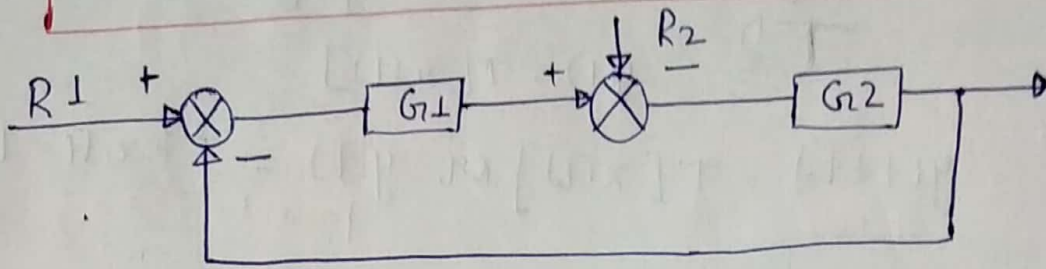
$$\begin{aligned} C_0 &= \frac{1}{1 + K_p} \\ C_1 &= \frac{1}{K_v} \\ C_2 &= \frac{1}{K_a} \end{aligned}$$

* Taylor's Series

$$C_k = \lim_{s \rightarrow 0} \frac{d^k F(s)}{ds^k} \quad k = 0, 1, 2, 3, \dots$$

$$C_0 = \lim_{s \rightarrow 0} F(s) = \lim_{s \rightarrow 0} \frac{1}{1 + OLTF} = \frac{1}{1 + K_p}$$

Above method for C_{ss} are valid only for SISO



$$C_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s \cdot E(s)$$

$$C = G_2 E_2$$

$$E_2 = E_1 G_1 - R_2$$

$$E_1 = R_1 - C$$

$$= R_1 - G_2 E_2$$

$$E_1 = R_1 - G_2 [E_1 G_1 - R_2]$$

$$E_1 [1 + G_1 G_2] = R_1 + R_2 G_2$$

$$C_{ss} = \lim_{s \rightarrow 0} s \cdot E_1(s)$$

$$E_1 = \frac{R_1}{1 + G_1 G_2} + \frac{R_2 G_2}{1 + G_1 G_2}$$

$$C_{ss} = \lim_{s \rightarrow 0} s \left[\frac{s \cdot R_1(s)}{1 + G_1 G_2} + \frac{s \cdot R_2(s) \cdot G_2(s)}{1 + G_1(s) \cdot G_2(s)} \right] E_1(s)$$

* When control system has multiple ip then then express second error in the form of first as above.

* Transportation Delay (or) Lag phase (or) Dead time

T.D

F.B

$$y(t) = T[x(t)]$$

$$y(t+t_0) = T[x(t)] \text{ or } y(t) = T[x(t-t_0)]$$

future present

L.T

$$\frac{y(s)}{x(s)} = e^{-st_0}$$

F.T

$$\frac{y(j\omega)}{x(j\omega)} = e^{-j\omega t_0}$$

$$\frac{e^{-st_0}}{sT}$$

$$\frac{\omega_n^2 \cdot e^{-st_0}}{s(s+2\zeta\omega_n)}$$

$$e^{-st_0} = 1 - st_0 + \frac{(st_0)^2}{2!} + \dots$$

$$e^{-j\omega t_0}$$

$$|e^{-j\omega t_0}| = 1$$

$$\text{phase delay} = 57^\circ \omega t_0$$

There is no effect of T.D or LP on M_p , e_{ss} .

* Due to transportation delay M_p , e_{ss} remains unaffected but t_p , t_d , t_r , t_s increases due to which, system becomes slower and less stable.

* In case of huge transportation delay system may become unstable though -ive feedback is present.

$$G_c \cdot H(s) = \frac{3 \cdot \omega_n^2}{s(s+2)} \quad 2\zeta\omega_n$$

$$G_c \cdot h(s) = \frac{3}{s^2(s+2)}$$

$$q(s) = 1 + \frac{3}{s^2(s+2)} = 0$$

$$s_{1,2,3} = -2.48, \pm j1.3$$

$\therefore$ unstable.

$$q(s) = 1 + \frac{3}{s(s+2)} = 0$$

$$q(s) = s^2 + 2s + 3 = 0$$

$$s_{1,2} = -1 \pm j\sqrt{2}, \text{ stable}$$

$$G_1 H(s) = \frac{3 \cdot e^{-s(1)}}{s^2(s+2)}$$

$$q(s) = 1 + \frac{3(1-2s)}{s(s+2)} = 0$$

$$q(s) = s^2 - 4s + 3 = 0$$

$$s_{1,2} = 1, 3$$

unstable.

$$G_2 H(s) = \frac{3000 \cdot e^{-s(0.002)}}{s(s+2)}$$

$$q(s) = 1 + \frac{3000(1-0.002s)}{s(s+2)} = 0$$

$$q(s) = s^2 - 4s + 3000 = 0$$

$s = +ive$ unstable.

* Forms of Transfer function:

$\downarrow$ pole-zero $\swarrow$ $\nearrow$ OL gain $\searrow$ OL DC gain $\downarrow$ Time constant form

$$G_1 H(s) = \frac{10}{s+2} = \frac{5}{\frac{1}{2}s+1} \xrightarrow{T_{OL}}$$

$$G_2 H(s) = \frac{100}{s^2(0.15s+1)(0.02s+1)} \xrightarrow{T_{OL}, T_{2,OL}}$$

* Time constant form of the transfer function is that in which all the finite zeros and all the finite poles are having '+1'.

* A zero or a pole is said to be finite if it is located neither at origin nor at infinity in s-plane.

* By changing type of T.F. = $\frac{2}{(\frac{1}{5}s+1)(s^2+s+1)}$

System DC gain value doesn't change but notation changes.

$$s^2 + s + 1 = 0$$

$$s = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$$

$$T_{2,3} = 2$$

Time constant form is used in Dominant Root Concept and Bode plot in Control System

T.f = $\frac{1}{1+ST}$ → CLDC gain $G.H(s) = \frac{1}{sT} = \frac{1/T}{s}$ → K_v
 T_{cl}
 $e_{ss} = \frac{1}{K_v} = T$
 type = 1

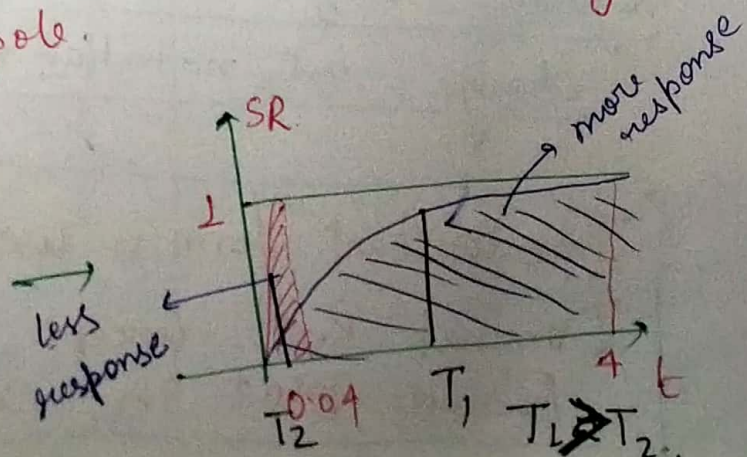
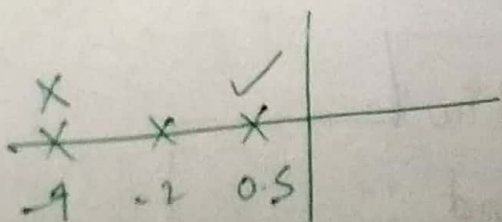
T.F = $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$
 $= \frac{1}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta\frac{s}{\omega_n} + 1}$

$G.H(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$
 $= \frac{\omega_n}{2\zeta} = K_v$
 type = 1.

* Dominant or Significant root:-

* Among finite zeros or poles of system. The pole closed to origin is called dominant pole. If the distance of location of other poles w.r.t to the location of dominant pole is greater or equal to 5 times then they are called ordinary or insignificant pole.

* If the distance is less than 5 times then they are also dominant pole.



The value of DC gain must be same after Dominant Concept
 Before Removing insignificant pole, first
 convert T.F in time constt form.

* Dominant root Concept is applicable in time constt form.

In higher order system always 2nd order poles dominate

* Sensitivity :

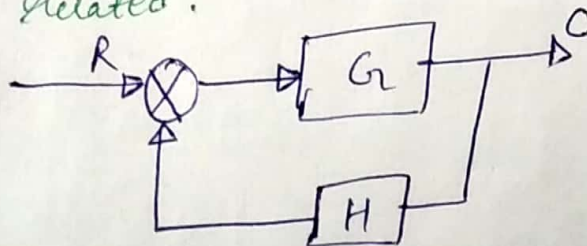
$$S_K^T = \frac{\frac{\delta T}{T}}{\frac{\delta K}{K}} = \left[\frac{K}{T} \cdot \frac{\delta T}{\delta K} \right]$$

T → function of

K → parameter

* Sensitivity is quantitative measure of amount of change in output due to change in the input or system parameters.

* Sensitivity and stability are inversely related.



$$T_1 = G_1$$

$$T_2 = \frac{G_1}{1 + G_1 H}$$

$$S_{G_1}^{T_1} = \frac{G_1}{T_1} \cdot \frac{\delta T_1}{\delta G_1}$$

$$= \frac{G_1}{G_1} \cdot \frac{\delta G_1}{\delta G_1}$$

$$= 1$$

$$S_{G_1}^{T_1} = 100\%$$

$$S_{G_1}^{T_2} = \frac{G_1}{T_2} \cdot \frac{\delta T_2}{\delta G_1}$$

$$= \frac{G_1}{G_1(1 + G_1 H)} \cdot \frac{\delta}{\delta G_1} \left[\frac{G_1}{1 + G_1 H} \right]$$

$$= (1 + G_1 H) \left[\frac{1 + G_1 H - G_1 H}{(1 + G_1 H)^2} \right]$$

$$= \frac{1}{1 + G_1 H}$$

* for +ive feedback

$$\boxed{S_H^{T_2} = \frac{1}{1 - GH}} \quad \triangleright 100\%$$

* -ive feedback decreases both gain and sensitivity by factor $\frac{1}{1+GH}$ hence increases stability compared to +ive feedback.

(for -ive feedback) $S_H^{T_2} = \frac{H}{T_2} \cdot \frac{S_H^{T_2}}{S_H}$

$$= \frac{H}{G(1+GH)} \cdot \frac{S}{S_H} \left\{ \frac{G_2}{1+GH} \right\}$$

$$= \frac{H \cdot (1+GH)}{G_2} \cdot \frac{S}{S_H} \frac{(GH+1) - G_2 L}{(1+GH)^2}$$

$$= \frac{-GH}{1+GH}$$

for +ive feedback

$$\boxed{S_H^{T_2} = \frac{+GH}{1-GH}}$$

$$\boxed{|S_H^{T_2}| > |S_G^{T_2}|}$$

* In control system plant has to be less sensitive but error detector, controller and sensor must be highly sensitive.

* we can calculate ^{natural} frequency from OL gain but not from CL gain

$$\omega_n = \sqrt{K}$$

$$\phi \propto \frac{1}{\sqrt{K}}$$

$K=0$ gain
for 2nd order

for 3rd order

$$\omega_n \propto \sqrt{K}$$

$$\phi \propto \frac{1}{\sqrt{K}}$$

Chapter-3

* Time Domain Stability Analysis:

* Concept of Stability:

↳ A system to be stable ^{if} all the roots of characteristic equation i.e. all the closed loop poles must be located in left side of S-plane irrespective of location of open loop poles.

- ①. If all the roots of $q(s)$ are having -ve real part (stable)
- ②. If any of the roots of $q(s)$ is having +ve real part $\Rightarrow$ unstable.
- ③. If any of the roots of $q(s)$ is having 0 real part $\Rightarrow$ while remaining roots are -ve.
(Marginal stable / critically stable)

1 $\rightarrow$ (c)

2 $\rightarrow$ (a)

3 $\rightarrow$ (d)

4 $\rightarrow$ (b)

5 $\rightarrow$ (c)

6 $\rightarrow$ (d)

7 $\rightarrow$ (b)

8 $\rightarrow$ (a)

18 $\rightarrow$ (d)

19 $\rightarrow$ (d)

20 $\rightarrow$ 3:2

21 $\rightarrow$ 2:3

22 $\rightarrow$ 0

23 $\rightarrow$ 0

24 $\left[\begin{matrix} A=5 \\ K=10 \end{matrix} \right]$

9 $\rightarrow$ (a)

10 $\rightarrow$ (a)

11 $\rightarrow$ (a)

12 $\rightarrow$ (b)

13 $\rightarrow$ (i) (d)

14 $\rightarrow$ (a)

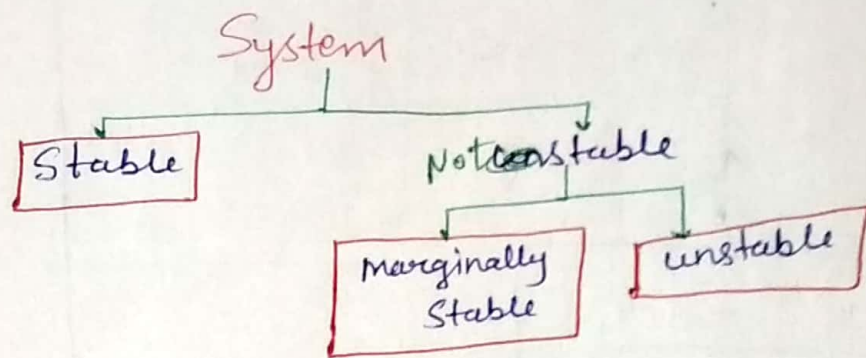
15 $\rightarrow$ (a)

16 $\rightarrow$ (b)

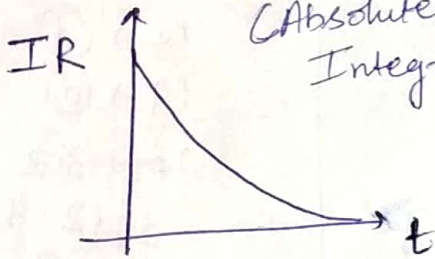
17 $\rightarrow$ (b)

25 $\rightarrow$ 1

26 $\rightarrow$ 10

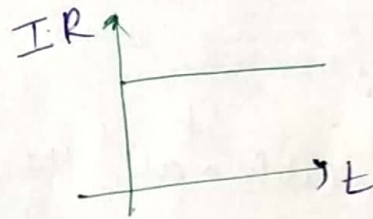
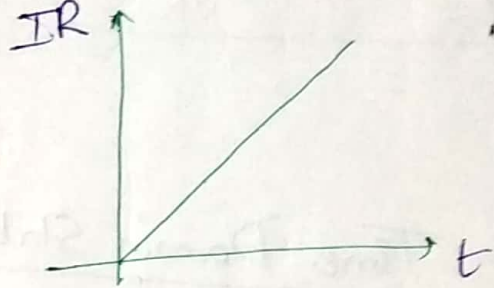


If IR is A.I (stable)
(Absolutely Integrable)



$IR \rightarrow 0$ as $t \rightarrow \infty$

If IR is NAI $\Rightarrow$ Not Stable

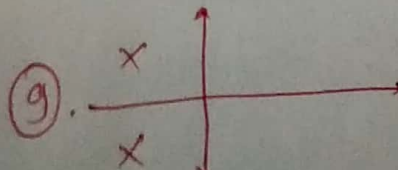
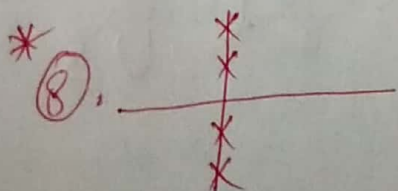
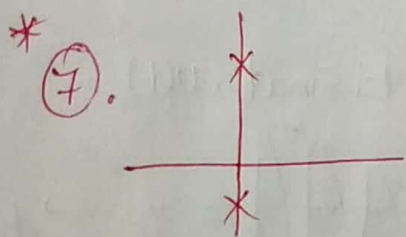
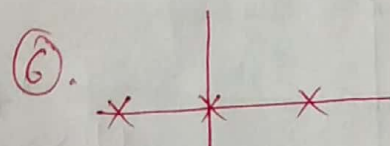
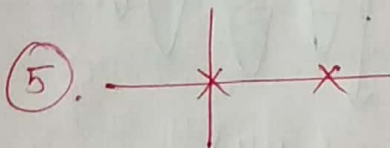
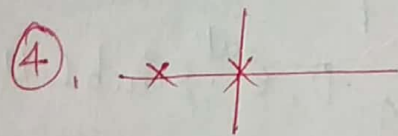
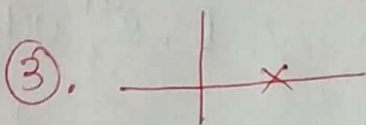
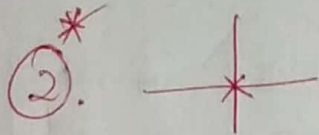
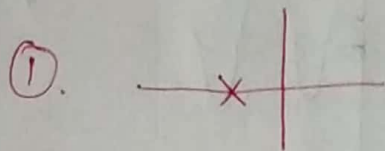


marginally stable.

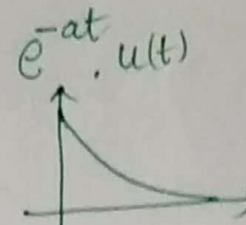
* Effects of location of roots of characteristic equation on impulse response.

* If multiple pole _{roots} on $(j\omega)$ axis i.e. origin then also system can be unstable.

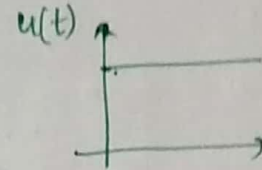
Location



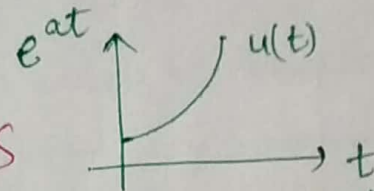
IR



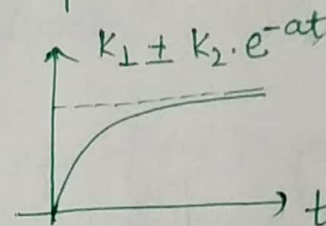
S



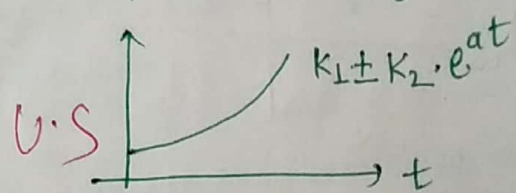
M.S



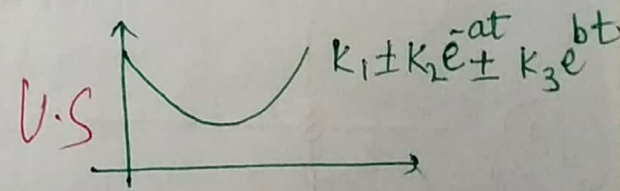
U.S



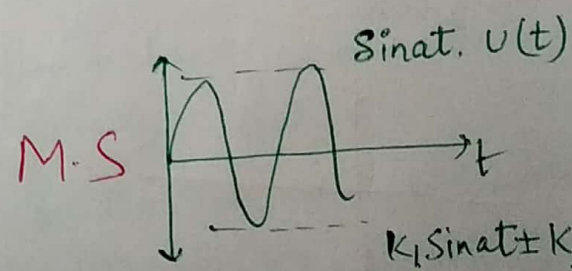
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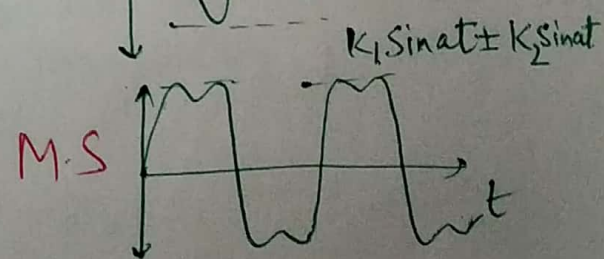
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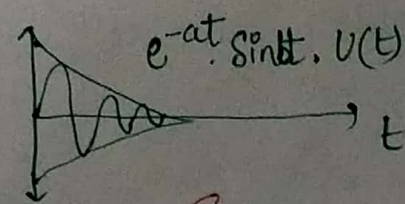
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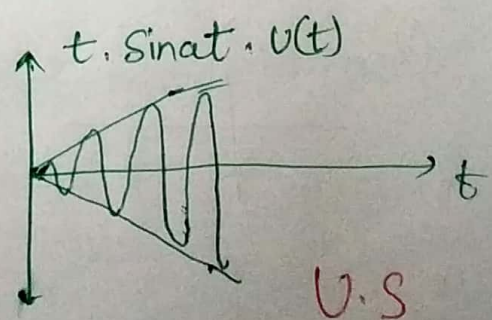
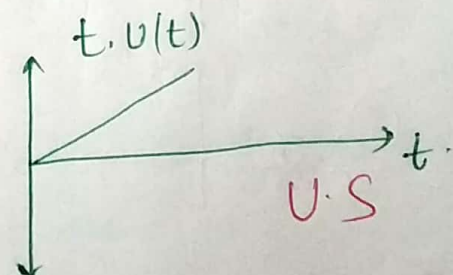
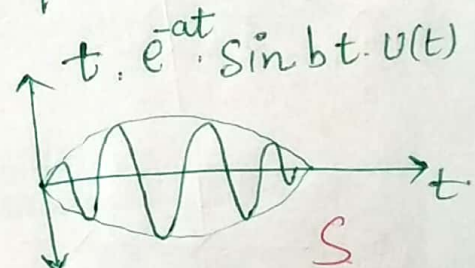
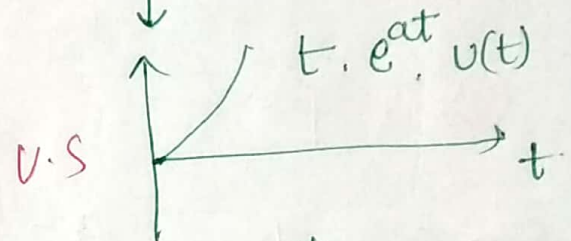
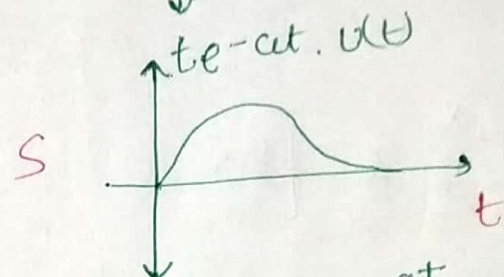
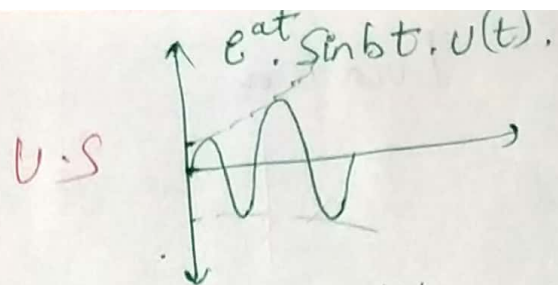
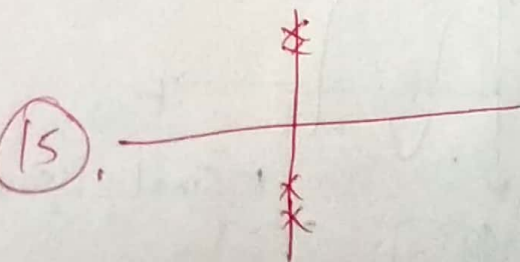
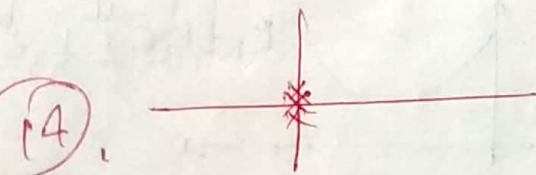
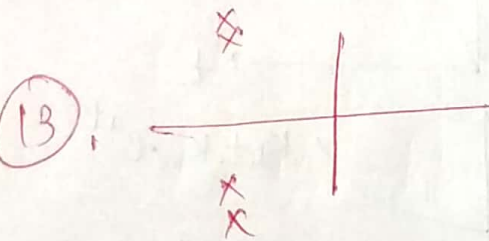
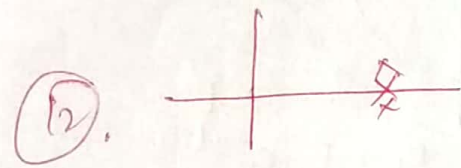
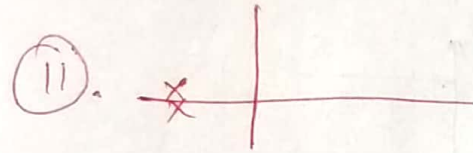
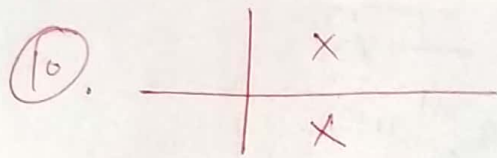
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M.S



S



* Types of Stability :

- 1). BIBO : ZSE } Response based.
 - 2). Asymptotic : ZIR }
 - 3). Absolute } system parameter based
 - 4). conditional }
 - 5). Relative.
- ↳ Response parameter based.

1). BIBO : If o/p of the system must be bounded if the i/p given to the system is bounded. This type of stability is based on zero state response.

2). Asymptotic : According to this stability o/p of the system must tends to become zero. This type of stability is based on zero i/p response of the system.

3, 4.) Absolute & Conditional : These types of stabilities are defined w.r.t a parameter of system.

* If the system is stable for all values of parameter then it is called absolutely stable. * If the system is stable only for few value of parameter then it is called conditionally stable.

* for the boundary value & limiting value conditionally stable system becomes marginally stable.

* Standard 1st & 2nd order system is Absolutely stable.

* System $\geq$ 3rd order are conditionally stable.

if stb $0 < K < \infty$ ③
if stb $K_1 < K < K_2$ ④

if $K < 10$ or $K > 20$; unstable
if $K = 10$ or $K = 20$; marginally stable.

* Relative: It is usually used to compare a stable system with itself or to compare group of systems.

* $M_p, t_s \rightarrow$ in time domain

* $M_x, G_m, P_m \rightarrow$ in frequency domain
can be used as measures of relative stability.

Never use dominant root concept in stability analysis.

* Routh - Hurwitz Stability Criterion:

Objective: To determine the stability of any order system without factorizing characteristic polynomial.

* Consider closed loop characteristic eqⁿ of n^{th} order:

$$q(s) = a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n = 0$$

$a_0, a_1, \dots, a_{n-1}, a_n \rightarrow$ Real constt coefficient.

R-H criteria can only be used in time Domain

* ~~if~~ R-H criteria is not applicable if the coefficients of characteristic equation are not real (in frequency domain).

* If the open loop transfer function has infinite length sequence.

$$q(s) = s^4 + s^3 + 2s^3 + 2s^2 + 1 = 0; \text{ stable.}$$

$$q(s) = s^4 - s^3 + 2s^2 + s + 1 = 0; \text{ unstable.}$$

$$q(s) = s^4 + s^2 + 1 = 0; \text{ Not stable}$$

M.S
U.S

* Necessary conditions:

- ① All the coefficient of characteristic equation must be present & must have same sign.
- ② If any coefficient of characteristic equation is missing then the system is either marginally stable or unstable.
But if there are sign changes then system is unstable.

* Sufficient conditions:

Routh Array:

s^n	a_0	a_2	a_4		a_{n-1}
s^{n-1}	a_1	a_3	a_5		a_n
s^{n-2}	b_1	b_2			
s^{n-3}	c_1	c_2			
$\vdots$	$\vdots$				
$\vdots$	$\vdots$				
s^0	z_1				

In case of 3rd order system only apply inner & outer coefficient product to determine the stability.

inner & outer P. (Stable).

- * All the elements of first column of Routh-array must have same sign. And no row should become zero completely.
- * If there are sign changes then the system is unstable & the no. of sign change indicate no. of closed loop poles in right side of S-plane.
- * If a row becomes zero completely then the system is marginally stable and if several row becomes zero completely, then the system is unstable.

NOTE:

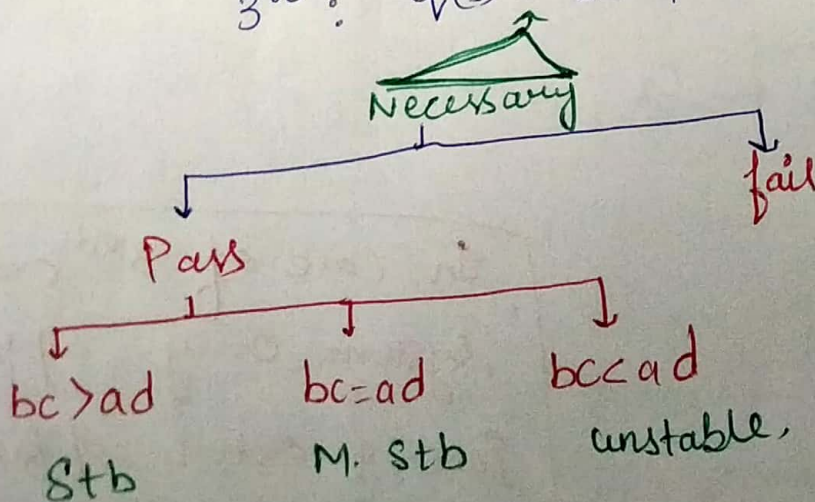
Roots of the characteristics equation can lie on jw axis of S-plane only if a row become zero completely but vice-versa need not be true.

1st: $q(s) = as + b = 0$

2nd: $q(s) = as^2 + bs + c = 0$

3rd: $q(s) = as^3 + bs^2 + cs + d$

} Necessary = Sufficient



Q. $q(s) = s^3 + 3s^2 + s + 5 = 0$

s^3	1	1	
s^2	3	5	
s^1	$-2/3$		
s^0	5		

unstable
(left) $L=1$
 $Im = 0$
 $R = \frac{2}{3}$
(right)

Q. $q(s) = s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

s^4	1	18	5
s^3	8	16	
s^2	16	5	
s^1	$18 \cdot 5$		
s^0	5		

Stable
 $L=4$
 $Im=0$
 $R = \frac{0}{4}$

$0 < K < \infty$ $-\infty < K < 0$
-ive feedback +ive feedback

Q. $s^4 + 5s^3 + 5s^2 + 4s + K = 0$ $\rightarrow$ OL gain

- find the range of values of K for the system to be stable.
- find the value of K to produce sustained oscillations in the system also find frequency of oscillations.

(i) $K > 0$
[to pass necessary condition]

s^4	1	5	K
s^3	5	4	
s^2	$+21/5$	K	
s^1	$\frac{84/5 - 5K}{21/5}$		
s^0	K		

> 0 and $\therefore 0 < K < \frac{84}{25}$

Conditionally stable.

if $K=0$ or $K=\frac{84}{25}$; marginally stable

if $K < 0$ or $K > \frac{84}{25}$; unstable

$s^2 + [2\zeta\omega_n s] + \omega_n^2 = 0$

$s^2 + \omega_n^2 = 0$

$\frac{21}{5}s^2 + \frac{84}{25} = 0 \Rightarrow s = \pm j\frac{2}{\sqrt{5}} = \omega$

$$q(s) = s^4 + 2s^3 + 3s^2 + 2s + k = 0$$

s^4	1	3	1	(i) $k > 0$
s^3	2	2		$\neq 2 - k > 0$
s^2	2	k		$k \neq 2$
s^1	$2 - k$			
s^0	k			$0 < k < 2$

(ii) for oscillations $\phi = 0$, $s'(\text{crit}) \downarrow 0$
 $\therefore 2 - k = 0, \Rightarrow k = 2$.

$$\omega_c = 1 \text{ rad/s}$$

$\omega \in [0, \infty)$

~~$2s^2 + 4k^2 = 0$~~
 ~~$2s^2 + k^2 = 0$~~
 ~~$s = \pm i\omega$~~

Q $G \cdot H(s) = \frac{k(s+3)}{(s+8)^2}$

$$q(s) = 1 + \frac{k(s+3)}{(s+8)^2} = 0 \Rightarrow (s+8)^2 + k(s+3) = 0$$

$$q(s) = s^2 + 64 + 16s + 3k + k \cdot s = 0$$

$$q(s) = s^2 + (16+k)s + (3k+64) = 0$$

s^2	1	$(3k+64)$
s^1	$16+k$	
s^0		

No need for
RH in 2nd
Order.

$$3k + 64 > 0$$

$$k > -\frac{64}{3}$$

$$k > -16$$

$$\Rightarrow \boxed{-16 < k < \infty}$$

$$-16 < k < 0$$

+ive f.B

$$0 < k < \infty$$

-ive f.B.

(ii)

$$S(K+16) \Rightarrow K = -16$$

for 2nd order only: $\omega_n = \sqrt{16} = 4 \text{ rad/s}$

$$s^2 + (3K+64) = 0$$

$$s^2 + 3(-16) + 64 = 0 \Rightarrow s = \pm j(4) = \omega_n$$

Q

$$\frac{K(s+1)^2}{(s+2)^2} \quad q(s) = 1 + \frac{K(s+1)^2}{(s+2)^2} \Rightarrow (s+2)^2 + K(s+1)^2 = 0$$

$$q(s) = s^2 + 4s + 4 + Ks^2 + K + 2KS = 0$$

$$q(s) = s^2(1+K) + s(2K+4) + (4+K) = 0$$

(ii) for stability.

[all the coefficient sign should same]

$$1+K > 0$$

$$2K+4 > 0$$

$$4+K > 0$$

$$1+K < 0$$

$$2K+4 < 0$$

$$4+K < 0$$

$$K > -1, K > -2, K > -4$$

$$K > -4, K < \infty$$

$$K < -1$$

$$K < -2$$

$$K < -4$$

$$-\infty < K < -4$$

(ii)

for sustained oscillations,

$$S(2K+4) \Rightarrow \boxed{K \neq -2}$$

~~there are no real parts.~~

$\therefore K = -2$ is not same as boundary value condition hence system is not marginally stable hence NO oscillatⁿ condition.

NOTE $\uparrow$

$\rightarrow K < -1$
 $-1 < K < 0$
 or
 $-\infty < K < -4$
 $\rightarrow 0 < K < \infty$
 M.S.C $K = -1$
 or $K = -4$

Q. $G H(s) = \frac{K(s^2+4)}{s(s^2+1)}$

$q(s) = s^3 + Ks^2 + s + 4K = 0, K > 0$

$bc = K$

$ad = 4K$

$\frac{bc < ad}{\text{unstable.}}$

* Correct Sequence of steps to stabilize a system:

- 1). Apply negative feedback.
- 2). Reduce open loop gain.
- 3). Introduce controller / compensator.

Q. $G H(s) = \frac{K(s+1)}{s^3 + as^2 + 2s + 1}$

$q(s) = s^3 + as^2 + 2s + 1 + K(s+1) = 0$

$q(s) = s^3 + as^2 + s(2+K) + K+1 = 0$

(i) Necessary conditions:-

$\boxed{a > 0}$; $K > -2$, $\boxed{K > -1}$

$bc = a(2+K)$

$ad = 1+K$

$a > \frac{1+K}{2+K} > 0$

K

a

2

0.6

2

0.8

3

0.6

3

0.8

$\boxed{-1 < K < \infty}$

→ for marginally stable.

* for marginally stable.

$$bc = ad \Rightarrow a(2+k) = 1+k.$$

$$a = \frac{1+k}{2+k}$$

$$as^2 + 1+k = 0$$

$$\left(\frac{1+k}{2+k}\right) \cdot s^2 + (1+k) = 0$$

$$s = \pm j\sqrt{k+2} \rightarrow \omega_n.$$

Shortcut method is only upto 3rd order

Q. $G.H(s) = \frac{k}{s+2}$, $q(s) = s+2+k=0$

(i) find the condition for closed loop to be located to the left of $s+5=0$ line.

~~Q(s) =~~ $q(s) = s+2+k=0$

$$s = -(k+2), \quad -(k+2) < -5.$$

$$s \rightarrow s-5.$$

$$q(s) = s-5+2+k=0$$

$$q(s) = s + (k-3) = 0$$

$$k-3 > 0 \Rightarrow k > 3.$$

$$\boxed{k > 3}$$

$$\boxed{3 < k < \infty}$$

Q. $q(s) = s^5 + 15s^4 + 85s^3 + 225s^2 + 274s + 120 = 0$

(i) Determine stability of system.

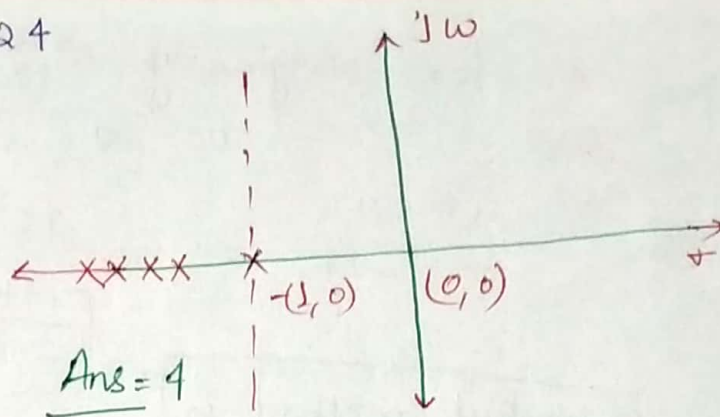
(ii) find the no. of roots of characteristic eqn that are located at left of

$s+1=0$ line.

$$q(s) = s^5 + 10s^4 + 35s^3 + 50s^2 + 24s = 0$$

by $s \rightarrow s-1$

s^4	1	35	24
s^3	10	50	
s^2	30	24	
s^1	42		
s^0	24		



Q Determine stability of following:-

$$q(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15 = 0$$

s^5	1	2	3
s^4	1	2	5
s^3	a	-12	
s^2	$+ \infty = \frac{2a+12}{a}$		
s^1	$\frac{-24a+44-15a^2}{2a+12} = -12$		
s^0	15		

$$\begin{aligned} L &= 3 \\ \text{Im} &= 0 \\ R &= \frac{2}{5} \end{aligned}$$

unstable

If any Row which is not zero completely if any element is zero or -ive then the system is unstable.

Reverse the coefficient to avoid limit calculations.

s^5	15	2	1
s^4	3	2	1
s^3	-8	-4	
s^2	$\frac{1}{2}$	1	
s^1	12		
s^0	1		

$$\begin{aligned} L &= 3 \\ \text{Im} &= 0 \\ R &= \frac{2}{5} \end{aligned}$$

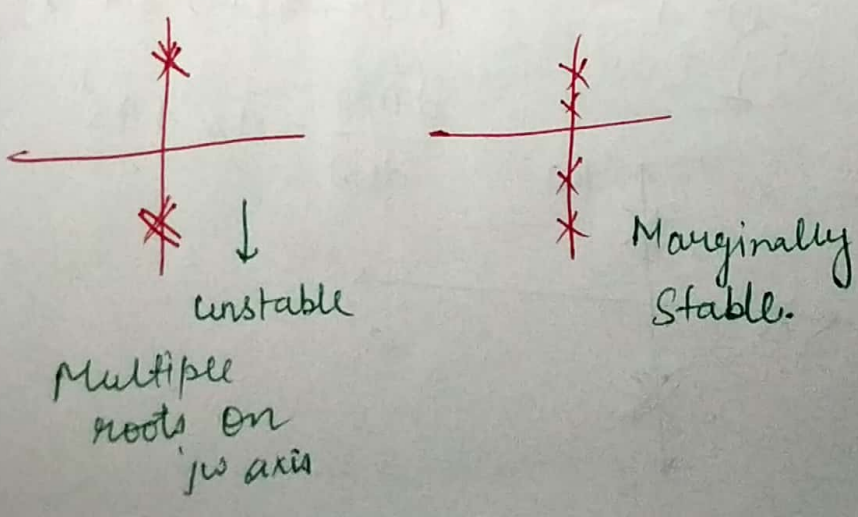
$Q(s) = s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$

s^6	1	8	20	16
s^5	2	12	16	
s^4	2	12	16	
s^3	0	8	0	24
s^2	6	16		
s^1	8	16		
s^0	16			

Asymmetric roots: s^6, s^5, s^4
 Symmetric roots: s^3, s^2, s^1, s^0

Auxiliary polynomial: $A(s) = 2s^4 + 12s^2 + 16$
 $\frac{Q(s)}{A(s)} = Q(s)$
 $\frac{d}{ds} A(s) = 8s^3 + 24s$

- * In the Routh-Arrary, one or more no. of odd rows become zero completely due to roots of the characteristic equation that are located symmetrically w.r.t origin.
- * In n-no. of roots [n Even] are symmetric to origin s^{n-1} row become zero completely.
- * If simple roots symmetric to origin then only one row become zero completely & if multiple roots is symmetric about origin then several row become zero.

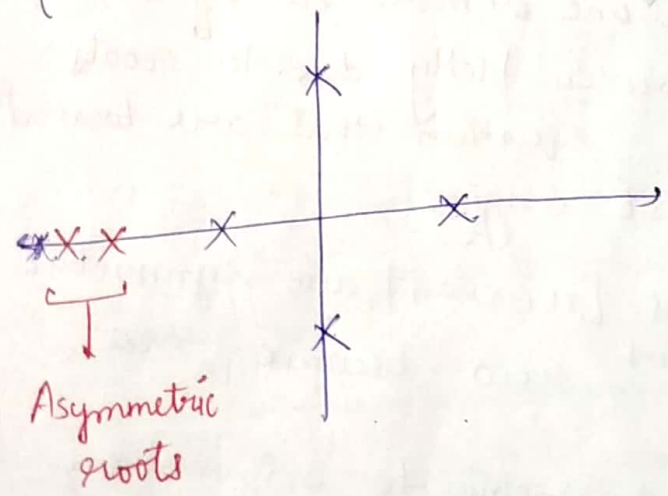


Q. $q(s) = s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8 = 0$
(unstable).

s^6	1	5	2	-8
s^5	1	3	-4	
s^4	2	6	-8	
s^3	2	12		
s^2	3	-8		
s^1	10/3			
s^0	-8			

$A(s) = 2s^4 + 6s^2 - 8$
 $\frac{dA(s)}{ds} = 8s^3 + 12s$

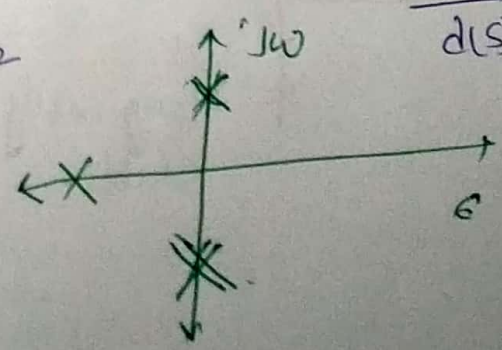
multiple row zero
↓
multiple roots
Single row zero.
↓
Simple roots.



Q. $q(s) = 2s^5 + s^4 + 4s^3 + 2s^2 + 2s + 1 = 0$

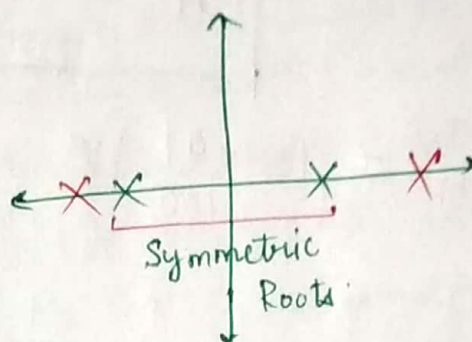
s^5	2	4	2
s^4	1	2	1
s^3	4	4	
s^2	1	1	
s	2		
s^0	1		

$A(s) = s^4 + 2s^2 + 1$
 $\frac{dA(s)}{ds} = 4s^3 + 4s$



Q. $q(s) = s^4 + s^3 - 3s^2 - s + 2 = 0$

$$\begin{cases} s^4 & 1 & -3 & 2 \\ s^3 & 1 & -1 \\ s^2 & -2 & -2 \\ s^1 & 0 & -4 \\ s^0 & 2 \end{cases}$$

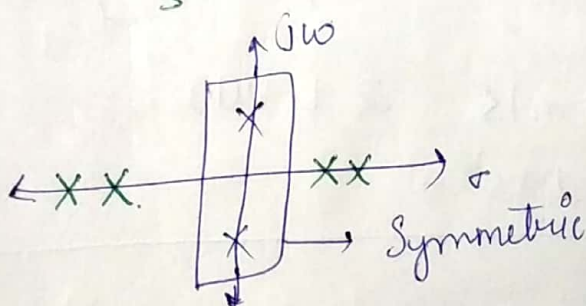


Q. $q(s) = s^6 + s^5 + 3s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$

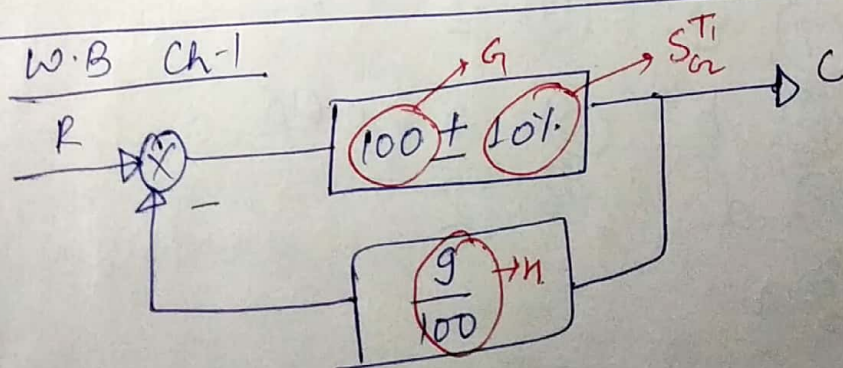
$\Delta = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$
above element

$$\begin{cases} s^6 & 1 & 3 & 3 & 1 \\ s^5 & 1 & 3 & 2 \\ s^4 & 0 & 1 \\ s^3 & - \\ s^2 & - \\ s^1 & - \\ s^0 & - \end{cases}$$

$$\begin{cases} s^6 & 1 & 3 & 3 & 1 \\ s^5 & 2 & 3 & 1 \\ s^4 & 3/2 & 5/2 & 1 \\ s^3 & -1/3 & -1/3 \\ s^2 & 1 & 1 \\ s^1 & 0 & 2 \\ s^0 & 1 \end{cases}$$



Q.



(a) $10 \pm 1\%$

A). CL gain = $\frac{100}{1 + 9n} = 10$

CL sensitivity = $\frac{10\%}{1 + 9n} = 1\%$

$$\frac{9}{100} \pm 10\%$$

$$S_{H}^{T_2} = G_{H1}, S_{G_1}^{T_2}$$

$$S_{H}^{T_2} = \left(100 \times \frac{9}{100}\right)\% = 9\%.$$

$$10 \pm 9\%$$

$$3). S_{G_1}^{T_2} = \frac{1}{1+G_{H1}} = \frac{1}{1+\frac{40}{s(s+4)}} = \left| \frac{s+4}{s+41} \right|_{s=0} \approx 0.1$$

$$6. C(0) = 0$$

4.)

$$\text{Ans 4: } S_{k}^{ess} = \frac{k}{e_{ss}}, \frac{\delta_{ess}}{\delta k} = -1$$

Ans 12: (b) (d)

$$2/s^2 \quad S_{a}^{ess} = \frac{a}{e_{ss}}, \frac{\delta_{ess}}{\delta a} = +1$$

$$5). e_{ss} \lim_{s \rightarrow 0} \left[\frac{s \cdot R(s)}{1+G_1(s) \cdot G_2(s)} + \frac{S T_D(s) \cdot G_2(s)}{1+G_1(s) \cdot G_2(s)} \right]$$

$$t_d(t) = 2 \text{ N-m/s} = 2 t \cdot v(t).$$

$$m s^2 =$$

W.B → 3rd

①. a

⑦. c

⑭. 2

②. d

⑧. b

⑮. $\alpha > 0.618$

③. c

⑨. d

⑯. 2

④. c

⑩. a

⑤. c

⑪. c

⑥. c

⑫. b

⑬. $\sqrt{6}$

* Root Locus Technique :-

— (W.R. Evans)

- * Root locus is graphical technique that deals with location of roots of the characteristics equation as a function of one of the system parameters. Usually open loop gain (K) varied from 0 to ∞ .
- * The Root locus obtained for $-ive$ & values of K . (+ive F.B system) is called Complementary or Inverse root locus.

$$G.H(s) = \frac{\overset{\text{OL gain}}{\underbrace{K}} \cdot \overset{\text{OL Z}}{\underbrace{(s+z_1) \dots}}}{\underbrace{s^x (s+p_1) \dots}_{\text{OL Pole}}}$$

$$q(s) = 1 + G.H(s) = \frac{s^x (s+p_1) \dots + K(s+z_1)}{s^x (s+p_1) \dots}$$

→ Z of $q(s)$

→ poles $q(s)$

Poles of $q(s)$ = OL Poles

$$q(s) = \frac{s^x (s+p_1) \dots + \underbrace{K(s+z_1)}_{\text{OL gain}}}{s^x (s+p_1) \dots} = 0$$

→ OL Pole
→ OL Z
→ CLP

$$K : 0 \rightarrow \infty$$

for $K=0 \Rightarrow$ OLP
 for $0 < K < \infty \Rightarrow$ CLP
 for $K=\infty \Rightarrow$ OLZ

* for constructⁿ of Root locus.

$$K \in [0, \infty)$$

* for study of Root locus
 $K \in (0, \infty)$

- * Root locus is the locus traced from open loop pole location to open loop zero location by varying open loop gain from zero to ∞ .

* Root locus is obtained using OL transfer function
it is locus of closed loop system.

$$OLP \longrightarrow OLZ$$

- (i) $P > Z$; strictly proper
(ii) $Z > P$; improper
(iii) $P = Z$; proper.

- (i) If $P > Z$ then $(P - Z)$ no. of zeros are at infinity.
(ii) If $Z > P$ then $(Z - P)$ no. of poles are at infinity.
(iii) If $P = Z$ then no zero or pole at ∞ .

* Properties of Root Locus Diagram:-

- ① Root locus diagram is always symmetric w.r.t real axis.
- ② No. of branches of root locus = No. of open loop poles or No. of open loop zeros [whichever is greater].
- ③ Root locus branch departs from open loop pole from $K = 0$ and arrives either at open loop zero or at ∞ for $K = \infty$.

④. Root locus exists at those points of real axis at the right of which total no. of open loop poles + zeros is odd.

⑤. $p-z$ no. of root locus branches arrives at infinity following asymptotes whose angles are given by:-

$$\phi_A = \frac{(2k+1) \cdot 180^\circ}{p-z}$$

$$k = 0, 1, 2, \dots, (p-z-1)$$

$p-z$	ϕ_A
0	undefined
1	180°
2	$90^\circ, 270^\circ$
3	$60^\circ, 180^\circ, 300^\circ$
4	$45^\circ, 135^\circ, 225^\circ, 315^\circ$

⑥. The point at which all the asymptotes meet at the ~~a~~ point at real axis known as centroid or center of gravity is given by:-

$$\sigma_A = \frac{\sum \text{real parts of poles} - \sum \text{real parts of zeros}}{p-z}$$

⑦. The point at which multiple roots of the characteristic eqnⁿ is present, is known as breakaway or break in commonly known as Saddle point. Can be obtained by solution of $\frac{dk}{ds} = 0$.

⑧* The angle of departure of root locus branch from simple real pole and the angle of arrival of root locus branch at simple zero is either 0° or 180° .

⑨. The angle of departure of root locus branch from multiple real ~~roots~~ poles and angle of arrival of root locus branch at multiple real zeros is given by.

$$\phi_m = \frac{(2k+1)}{n} \cdot 180^\circ$$

$$k = 0, 1, \dots, (n-1).$$

n = No. of multiple poles, zeros.

n	ϕ_m
2	$90^\circ, 270^\circ$
3	$60^\circ, 180^\circ, 300^\circ$
4	$45^\circ, 135^\circ, 225^\circ, 315^\circ$

⑩. The angle of departure of root locus branch from complex conjugate pair of poles both simple and multiple is given by

$$\phi_D = \pm [(2k+1)180^\circ + \phi]$$

$\xrightarrow{\text{complex}}$
 $\xrightarrow{\text{conjugate}}$

ϕ = net angle contribution at complex pole due to all other

k = Any integer value zeros & poles

- (11). The angle of arrival of the root locus branch at complex conjugate pair of zeros both simple and multiple is given by

$$\phi_{\text{Arrival}} = \pm \left[(2K+1)180^\circ - \phi \right]$$

$\xrightarrow{\text{complex}}$
 $\xrightarrow{\text{conjugate}}$

$$\phi_{\text{Arrival}} = \pm [180^\circ - \phi]$$

ϕ = Net angle contribution at the complex zero due to all other poles and zeros.

- (12). The point at which root locus branch intersects jw axis ~~at that~~ ^{value of} value of K at that point can be obtained from Routhen-Array.

- (13). The Value of the open loop gain at any point on the root locus is given by: -

Product of length of phasors drawn from all the open loop poles to that point

Product of length of phasors drawn from all the open loop zeros to that point.

- (14). A point S_0 chosen from any part of s-plane can lie on root locus of system if it satisfies the following condition.

* Magnitude Condition:

$$|G(s) \cdot H(s)| = 1$$

OR

* Phase Condition:

$$|G(s) \cdot H(s)| = \pm (2K+1) \cdot 180^\circ = \pm 180^\circ$$

$K = \text{any integer}$

* Properties of C.R. Locus Diagram:

①

②

③

* ④

~~odd~~ $\rightarrow$ even $\rightarrow 2K \text{ even}$

* ⑤

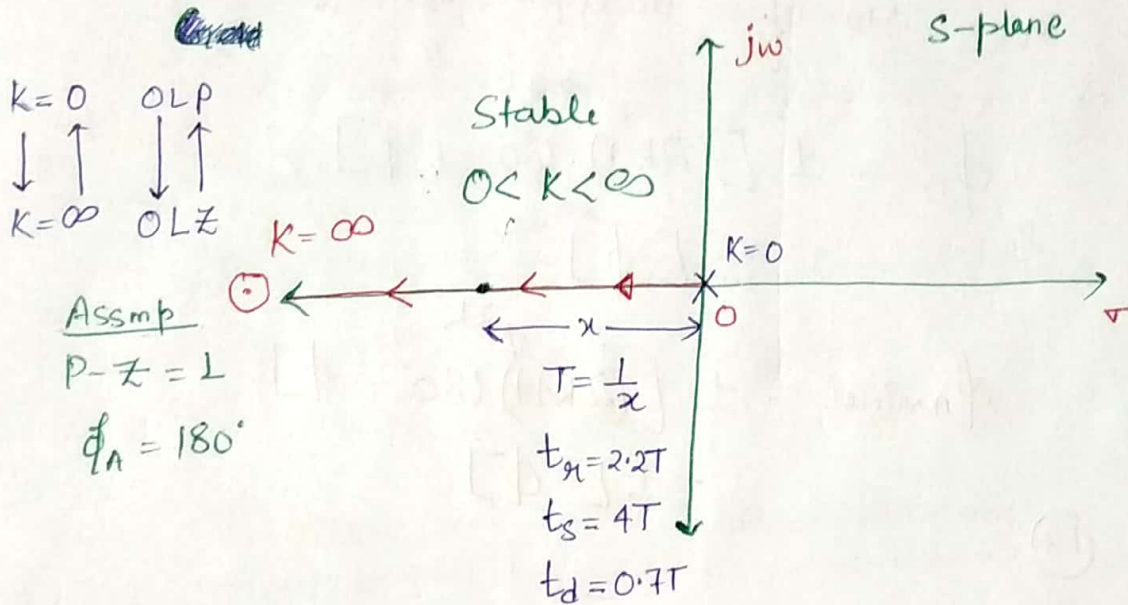
$$\phi_A = \frac{(2K+1) \cdot 180^\circ}{P-Z}$$

$P-Z$	ϕ_A
0	undefined
1	0°
2	$0^\circ, 180^\circ$
3	$0^\circ, 120^\circ, 240^\circ$
4	$0^\circ, 90^\circ, 180^\circ, 270^\circ$

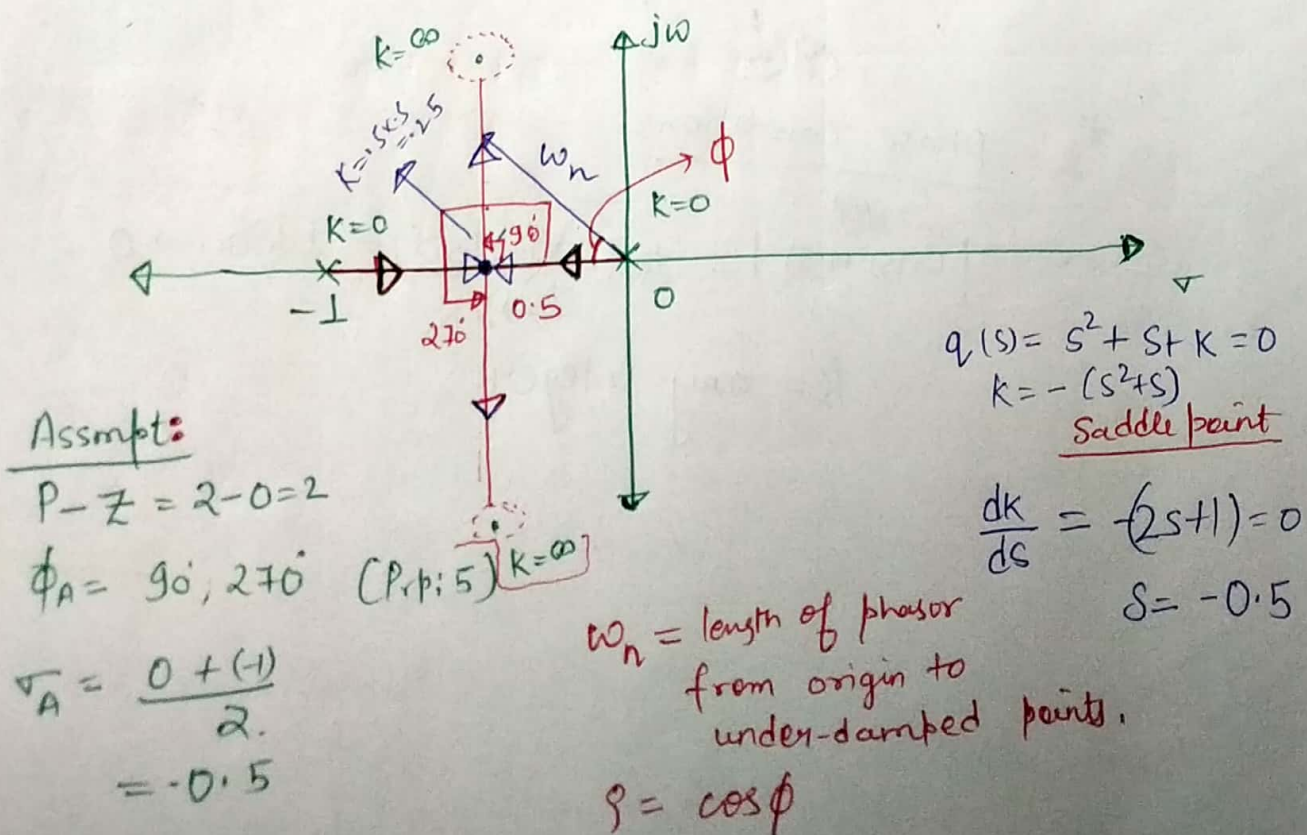
⑥, ⑦, ⑧

* Root Locus of Standard 1st & 2nd Order:

1st order: $G(s)H(s) = \frac{1}{sT} = \frac{\frac{1}{T}}{s} = \frac{K}{s}$



Second Order: $G(s)H(s) = \frac{\omega_n^2 K}{s(s+2\zeta\omega_n)} = \frac{K}{(s+1)s}$

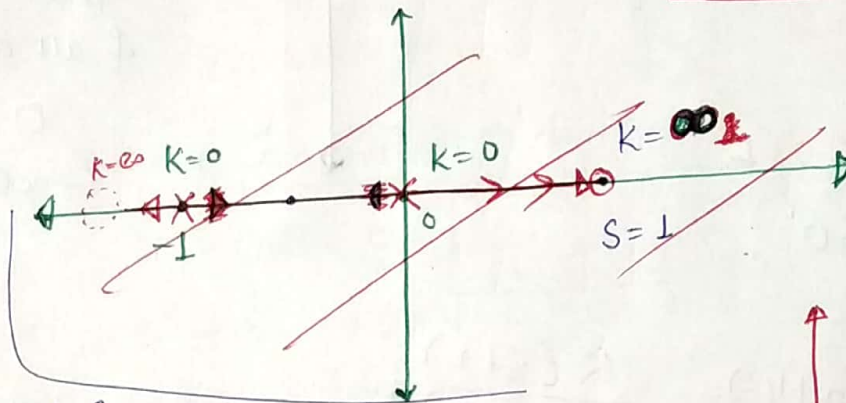


$k = 0.25$; critically damped.
 $0.25 < k < \infty$; under-damped.
 $0 < k < 0.25$; over-damped.

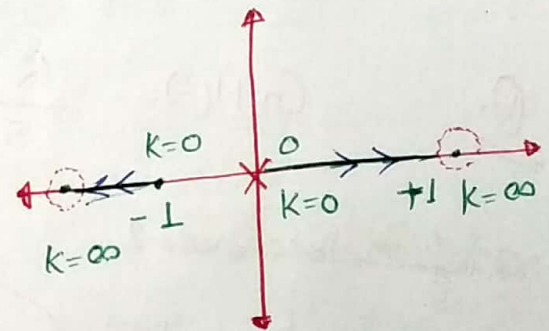
k at any point
 is determined by
 property: 13.

we can find k
 from $k = -(s^2 + s)$
 if s is Real only.

Q. $G.H(s) = \frac{k(s-1)}{s(s+1)}$



* Saddle point is obtained
 from the solution of $\frac{dk}{ds}$
 but solution of dk/ds need
 not to be saddle point.



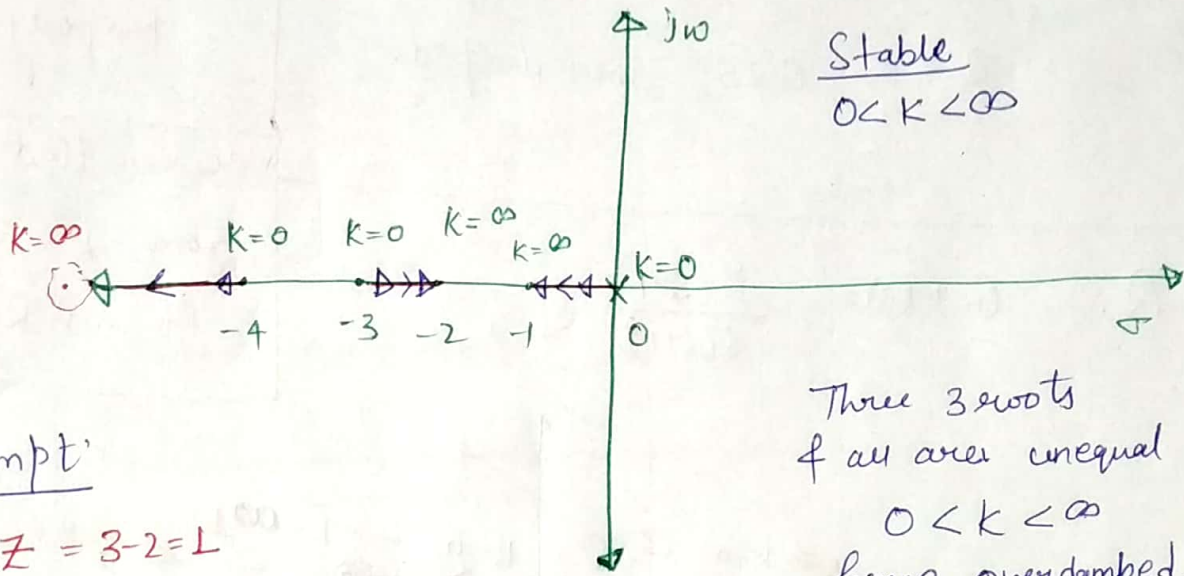
* for it to be saddle point it
 should lie on root locus.

unstable

$0 < k < \infty$

if a complete
 branch lie
 on right side
 then system
 is unstable.

Q. $G.H(s) = \frac{K(s+1)(s+2)}{s(s+3)(s+4)}$



Assmpt'

$P-Z = 3-2=1$

$\phi_A = 180^\circ$

Q. $G.H(s) = \frac{K(s+2)}{s(s+1)}$

~~No. of branches = 2~~

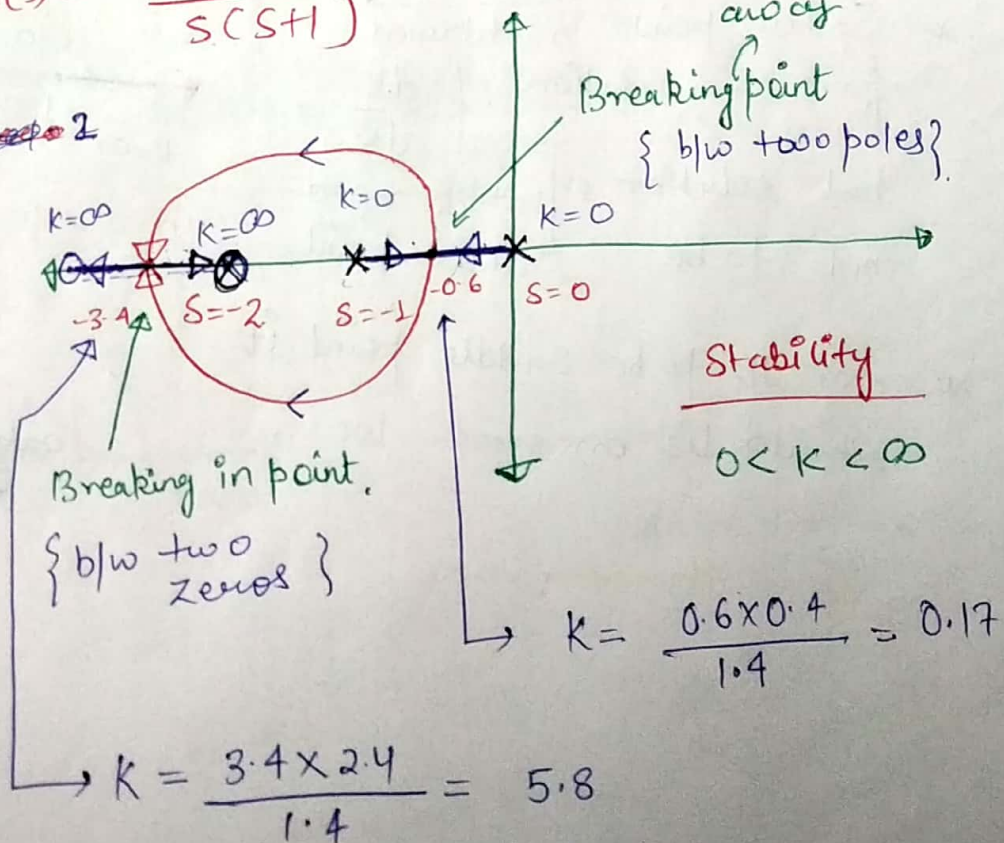
No. of Assp:

$P-Z = 1$

$\phi_A = 180^\circ$

$\frac{dK}{ds} = 0$

$s = -3.4, -0.6$



$0 < K < 0.17$; overdamped

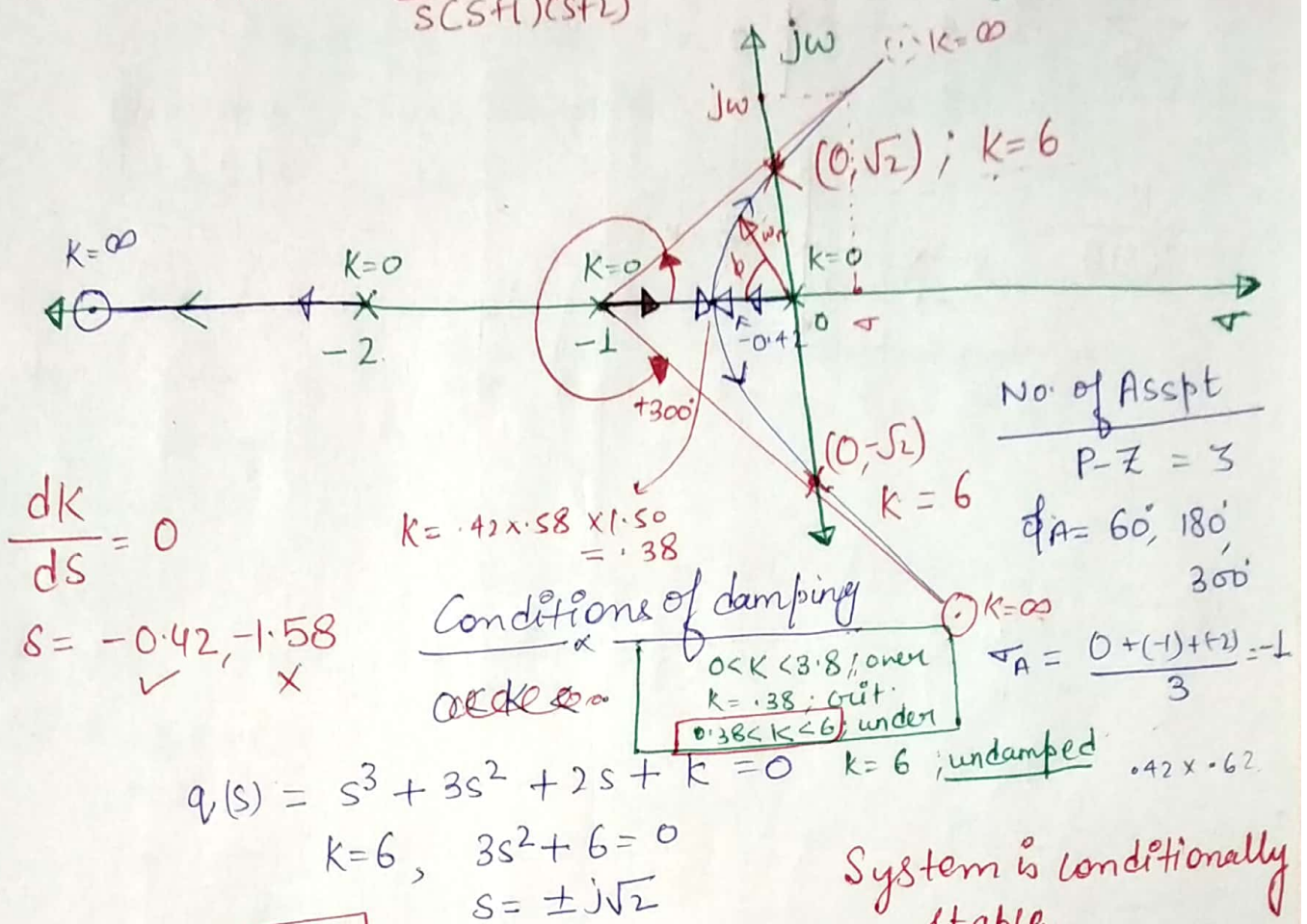
$K = 5.8, K = 0.17$; critically damped

$K > 5.8$; overdamped

~~Stable~~

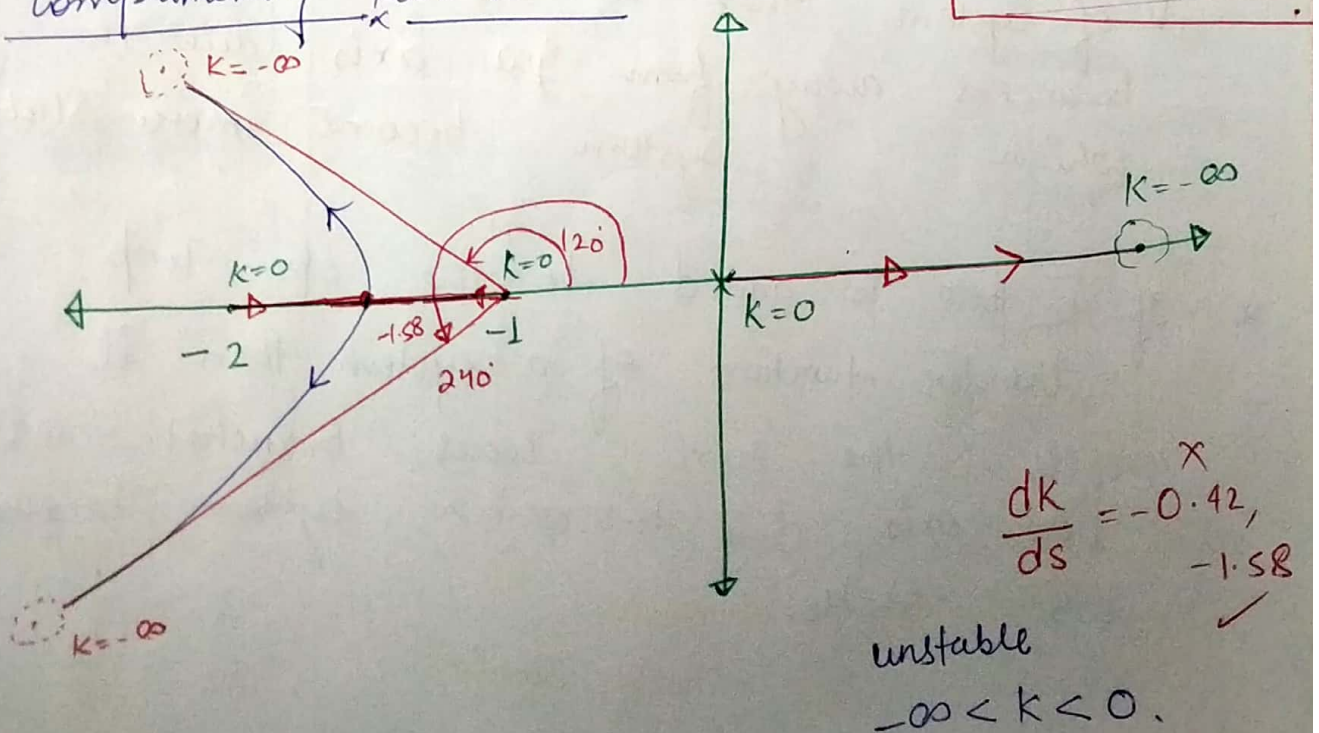
Q. $G.H(s) = \frac{K}{s(s+1)(s+2)}$

Obt Root locus diagram & complimentary Root locus diagram.

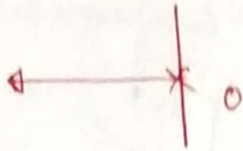


$\rho = \cos \phi$

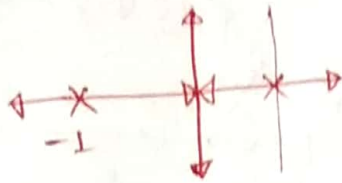
* Complimentary Root Locus:



$$\frac{K}{S}$$



$$\frac{K}{S(S+1)}$$



* If a zero is added to the open transfer function of system then it attracts root locus branches away from $j\omega$ axis due to which system becomes more stable.

* If a pole is added to the open loop transfer function of a system then it repels the root locus branches towards $j\omega$ axis. Due to which system becomes less stable.

* The zero or pole to be added must be such that it lies in left side of S-plane, such system is called minimum phase system.

* Zero is added alone but pole is added along the zero in control system design.

* $q(s) = s^2 + s(K+2) + 2(K+1) = 0$

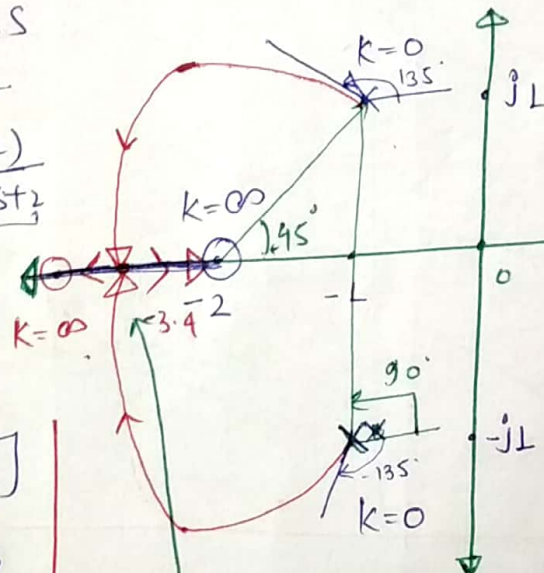
obt RL & CRLD.

$$\begin{aligned} K &> -1 \\ K &> -2 \\ K &> -2 \end{aligned}$$

$$q(s) = s^2 + s(K+2) + 2(K+1)$$

$$q(s) = 1 + \frac{k(s+2)}{s^2 + 2s + 2}$$

$G \cdot H(s)$



$$\phi_D = \pm [180^\circ + \phi]$$

~~$$\phi_D = \pm [180^\circ + \phi]$$~~

$$\phi = +45^\circ - [+90^\circ] = -45^\circ$$

$$\phi_D = \pm 135^\circ$$

Assmpt:

$$P - Z = 1$$

$$\phi_A = 180^\circ$$

$$k = \frac{2.6 \times 2.6}{1.4} = 4.8$$

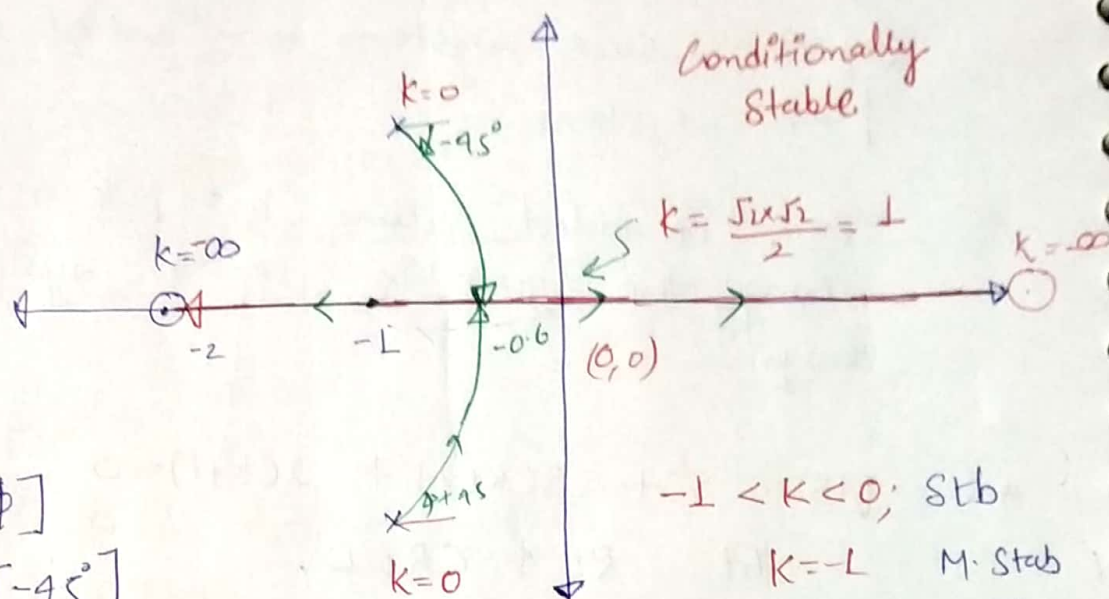
NOTE:

$$\begin{aligned} \left| \frac{Z}{P} \right| &= + |Z - P| \\ &= + |Z - P| \end{aligned}$$

$$\frac{dk}{ds} = 0$$

$$s = -3.4, -0.6$$

* Complimentary Root Locus:



$$\phi_D = \pm[\phi]$$

$$= \pm[-45^\circ]$$

$$= \mp 45^\circ$$

$$P-Z = L$$

$$\phi_A = 180^\circ - 0^\circ$$

$$\frac{dK}{ds} = 0$$

$$s = -3.4, -0.6$$

$$G_H(s) = \frac{K(s^2+4)}{s(s^2+1)}$$

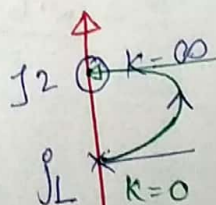
$$P-Z = 1; \phi_A = 180^\circ$$

$$\phi_D = \pm[180^\circ + \phi]$$

$$\phi = +(-90^\circ + 90^\circ)$$

$$-(+90^\circ + 90^\circ)$$

$$= -180^\circ$$



$$\phi_{Arrival} = \pm[180^\circ - \phi]$$

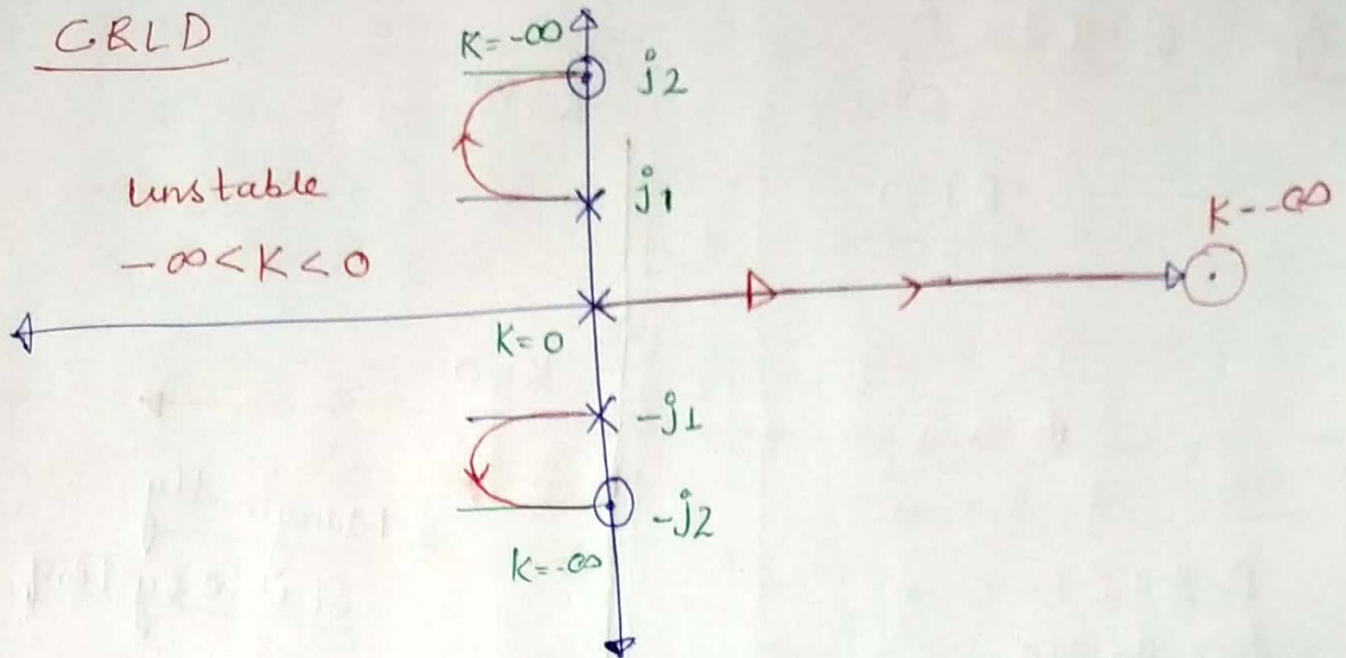
$$\phi = +90^\circ - (90^\circ + 90^\circ + 90^\circ)$$

$$= -180^\circ$$

$$\phi_A = 0^\circ$$

System is Ustable
 $0 < k < \infty$

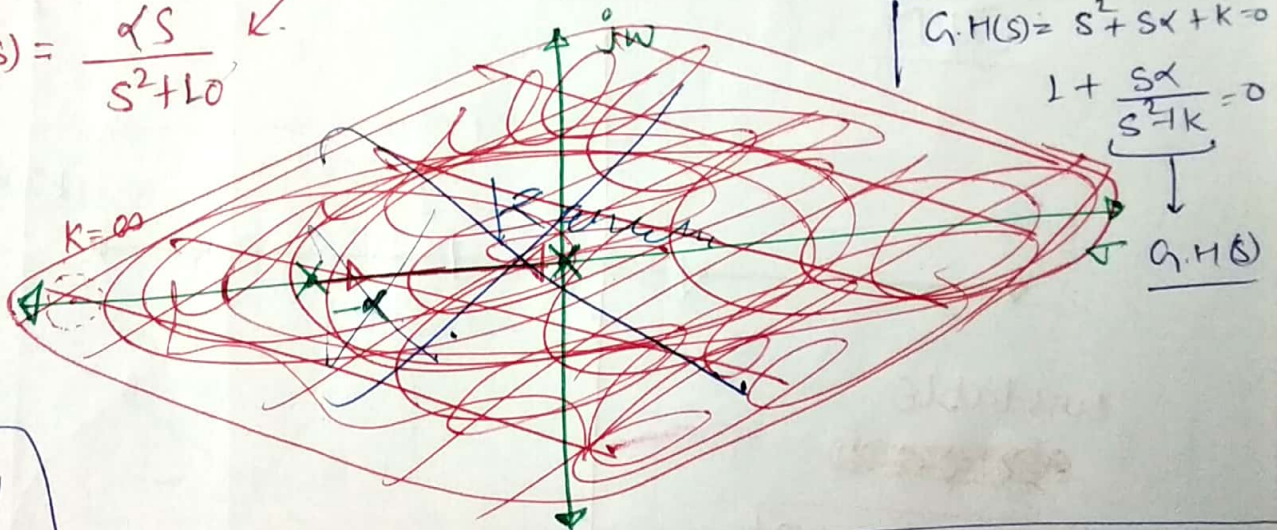
C.R.L.D



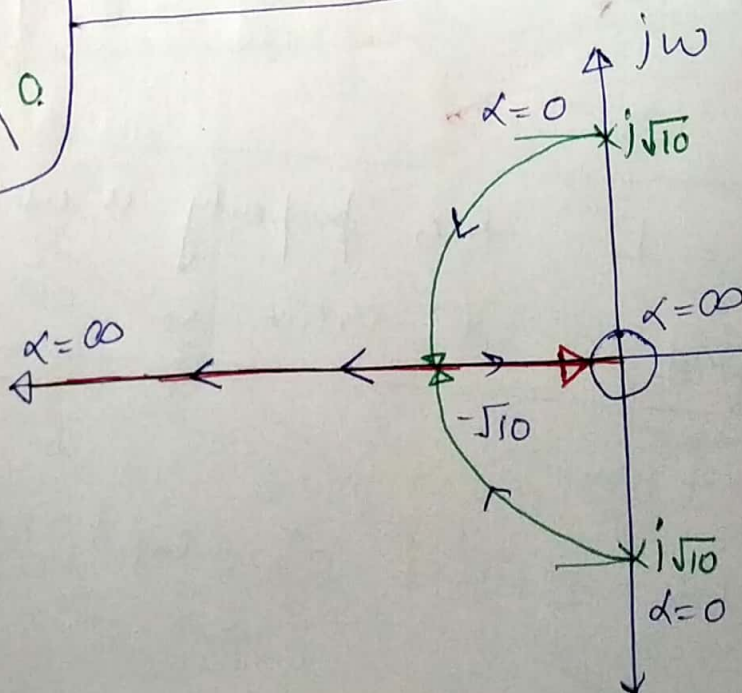
Q. $G.H(s) = \frac{K}{s(s+\alpha)}$ obtain RLD for $K=10; 0 < \alpha < \infty$.

$G.H(s) = \frac{\alpha s}{s^2 + 10}$

$G.H(s) = \frac{s^2 + s\alpha + K}{s^2 + 10}$
 $1 + \frac{s\alpha}{s^2 + 10} = 0$
 $\downarrow$
 $G.H(s)$



$\frac{dK}{ds} = 0$

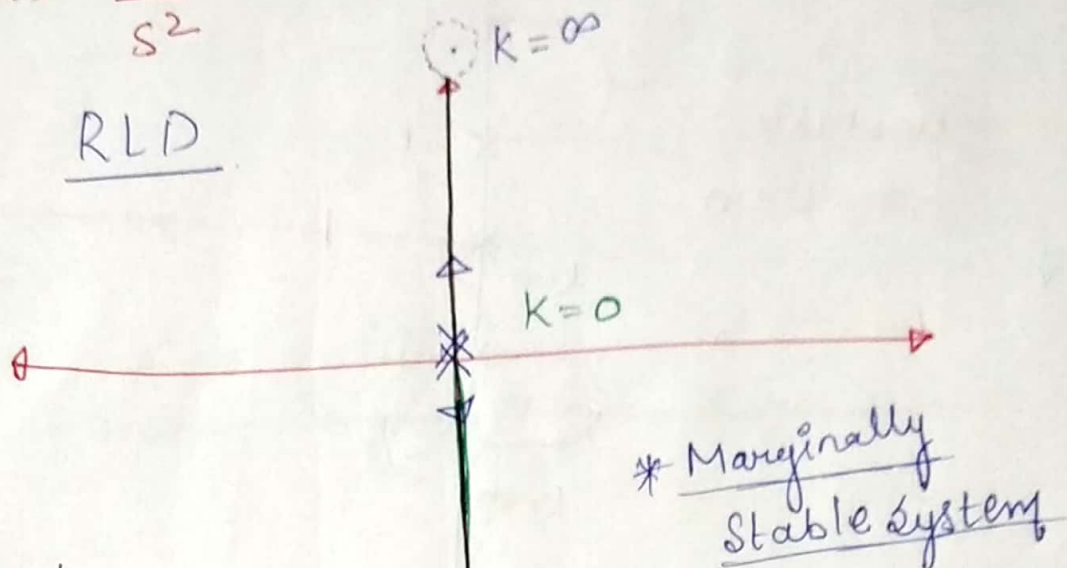


$\phi_D = \pm (180^\circ + \phi)$
 $\phi = +90^\circ - (+90^\circ) = 0$
 $\phi_D = \pm 180^\circ$
 $PZ = L \quad \phi_A = 180^\circ$

$\frac{d\alpha}{ds} = 0$
 $s = -\sqrt{10}, \sqrt{10}$

Q. $G.H(s) = \frac{K}{s^2}$

RLD

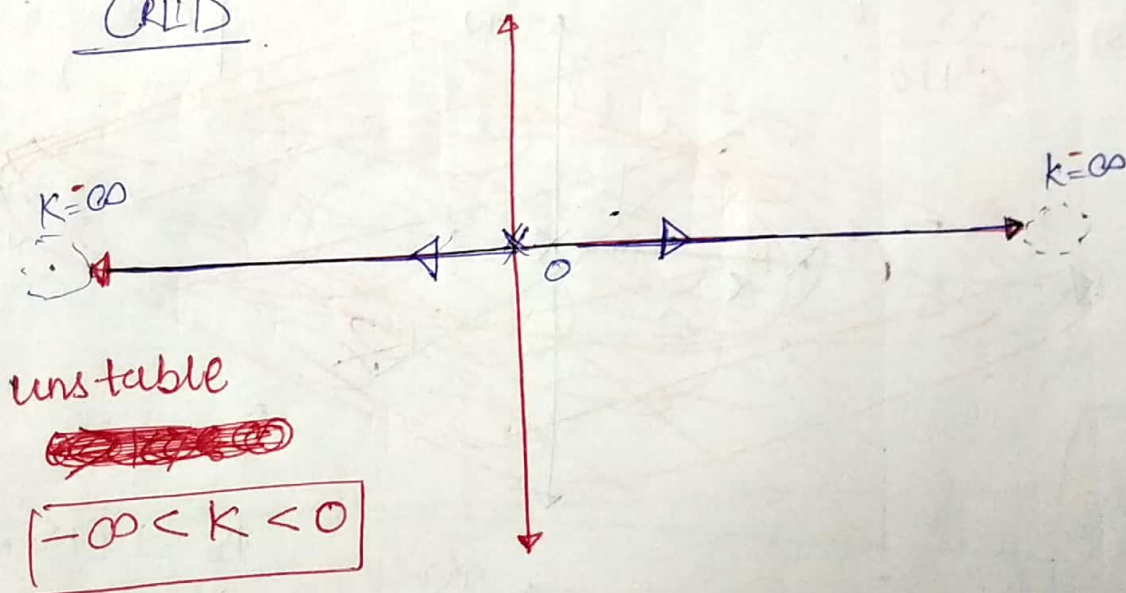


$P-Z=2$

$\phi_A = 90^\circ, 270^\circ$

$\sigma_A = \frac{0+0}{2} = 0$

CRLD



NOTE:

$|G(s).H(s)|_{s_0} = 1$ this property is used to find K at value s_0

$|G(s).H(s)|_{s_0} = \pm 180^\circ$

$\angle K/s^2 = \angle K - \angle (1+j)^2 = 0^\circ - 2 \tan^{-1}(\frac{1}{1}) = -90^\circ$

$$\angle \frac{K}{s^2} = \angle K - \angle (s^2) = 0^\circ - 2 \tan^{-1}(\frac{2}{0}) = -180^\circ \checkmark$$

$$\angle x = 0^\circ$$

$$\angle jy = +90^\circ \text{ or } -270^\circ$$

$$\angle -x = \pm 180^\circ$$

$$\angle -jy = -90^\circ \text{ or } +270^\circ$$

$$\text{I. } \angle x + jy = \tan^{-1}(\frac{y}{x})$$

$$\text{II. } \angle -x + jy = -\tan^{-1}(\frac{y}{x}) \pm 180^\circ$$

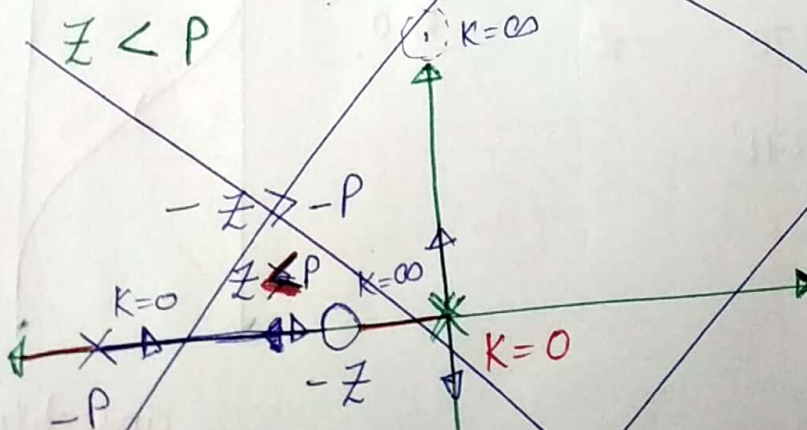
$$\text{III. } \angle -x - jy = \tan^{-1}(\frac{y}{x}) \pm 180^\circ$$

$$\text{IV. } \angle x - jy = -\tan^{-1}(\frac{y}{x})$$

M. Stb + Z ; Stable
M. Stb + P ; Unstb
Unst + Z = Can't say
Stb + P = Can't say

Q. $G.H(s) = \frac{K(s+Z)}{s^2(s+P)}$

$$Z < P$$

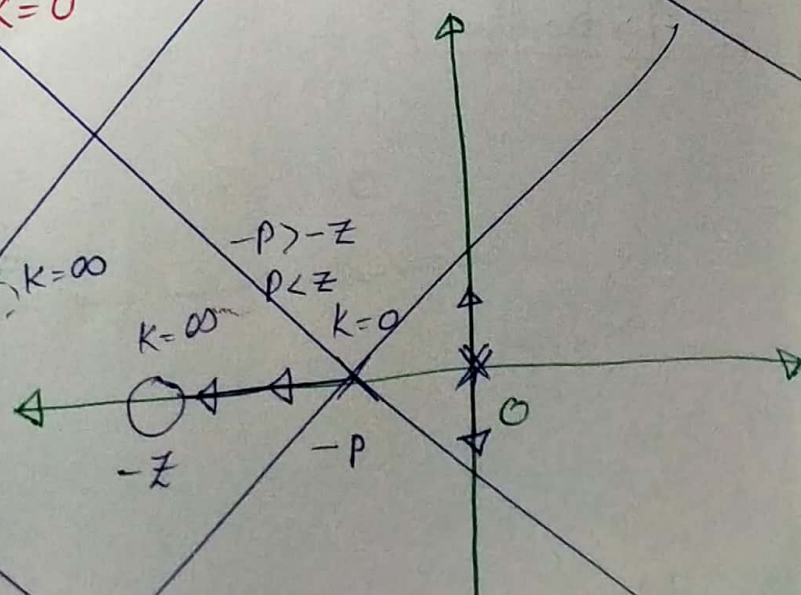


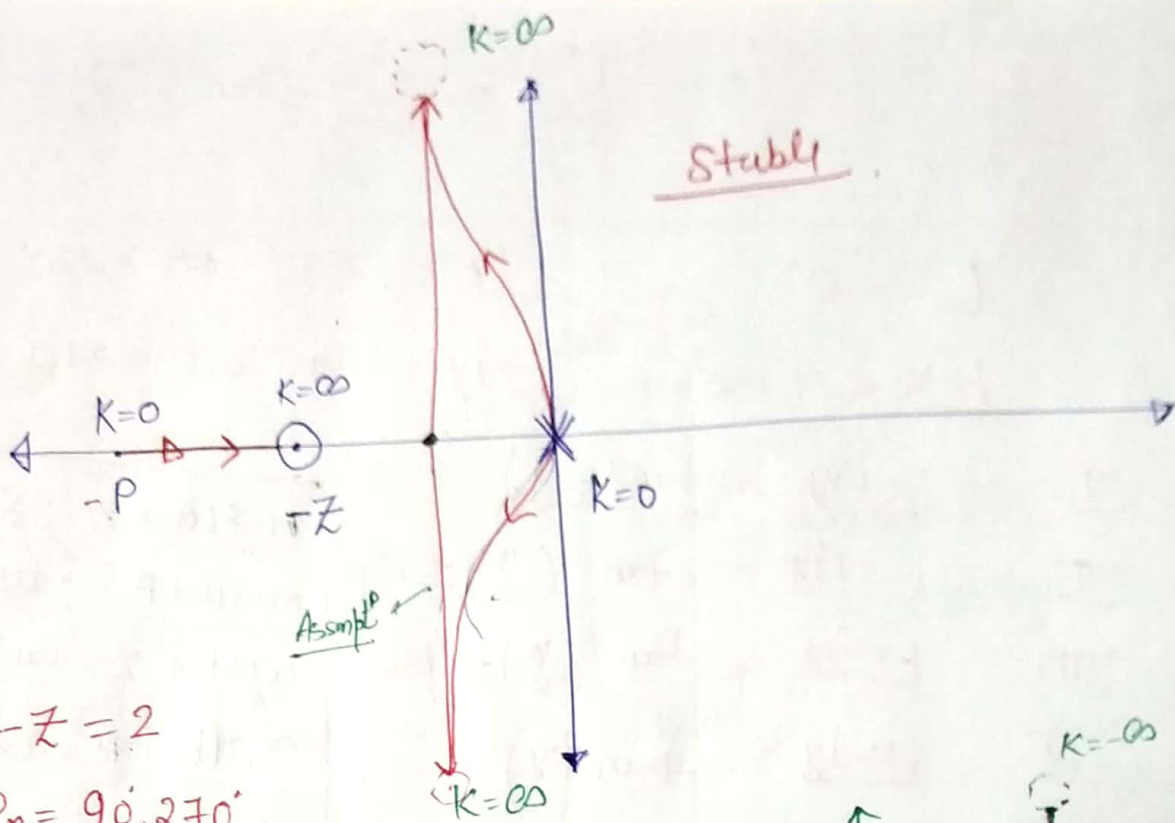
$$P - Z = 2$$

$$\phi_n = 90^\circ, 270^\circ$$

$$\sigma_n = \frac{0+0+(-P)-(-Z)}{2}$$

$$\frac{Z-P}{2} < 0$$





$$P - Z = 2$$

$$\phi_n = 90^\circ, 270^\circ$$

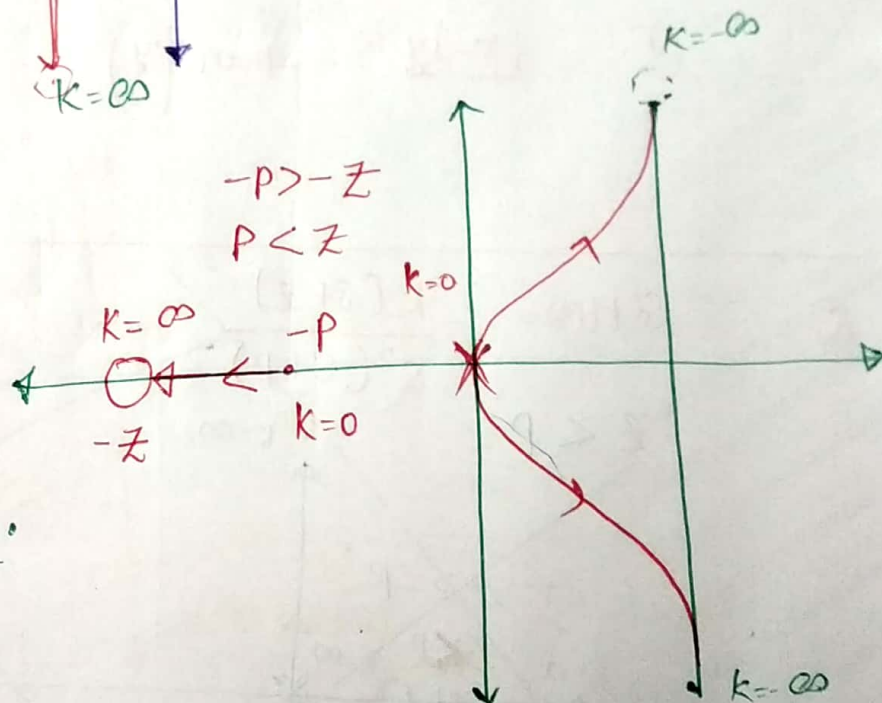
$$\sigma_A = \frac{Z - P}{2} > 0$$

$$< 0$$

$$K = 0 \Rightarrow \phi_D = 90^\circ, 270^\circ$$

$$K = 0^+ \Rightarrow \phi_D = 91^\circ, 271^\circ$$

$P < Z$
 $\rightarrow$ Dominant

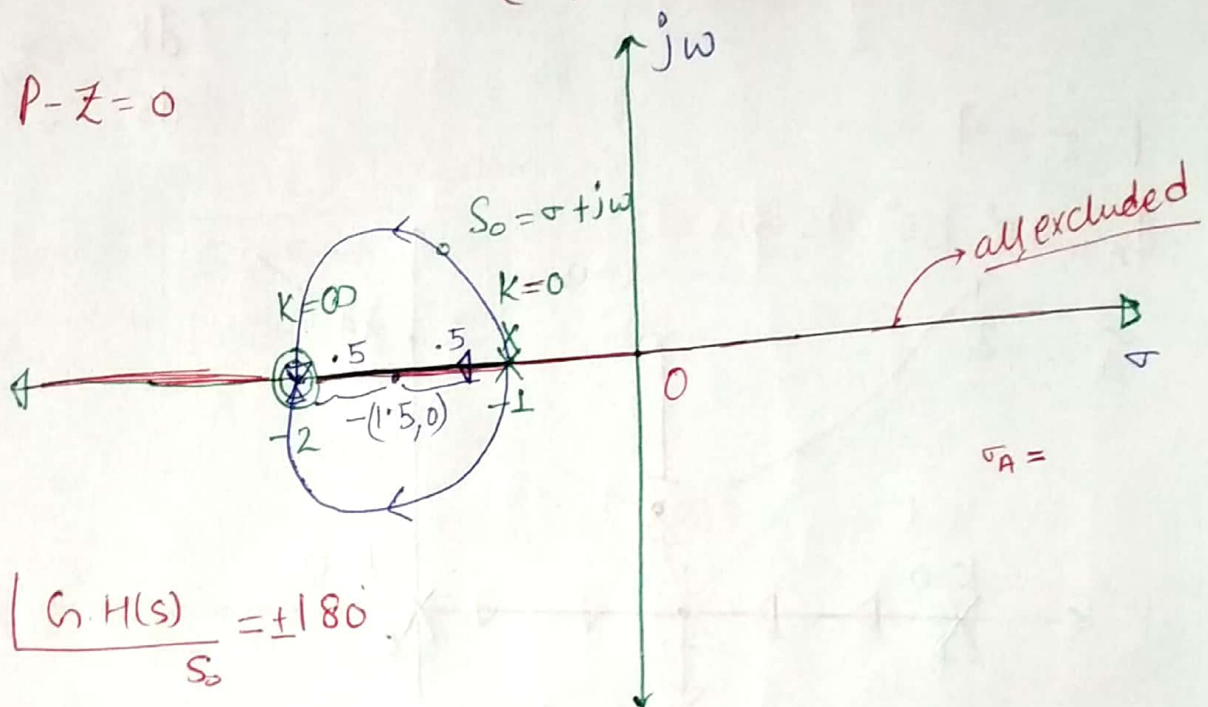


unstable

Q.

$$G \cdot H(s) = \frac{k(s+2)^2}{(s+1)^2}$$

$$P-Z=0$$



$$\angle \frac{G \cdot H(s)}{S_0} = \pm 180^\circ$$

$$\angle \frac{k(\sigma + jw + 2)^2}{(\sigma + jw + 1)^2} = 180^\circ$$

$$2 \tan^{-1}\left(\frac{w}{\sigma + 2}\right) - 2 \tan^{-1}\left(\frac{w}{\sigma + 1}\right) = 90^\circ$$

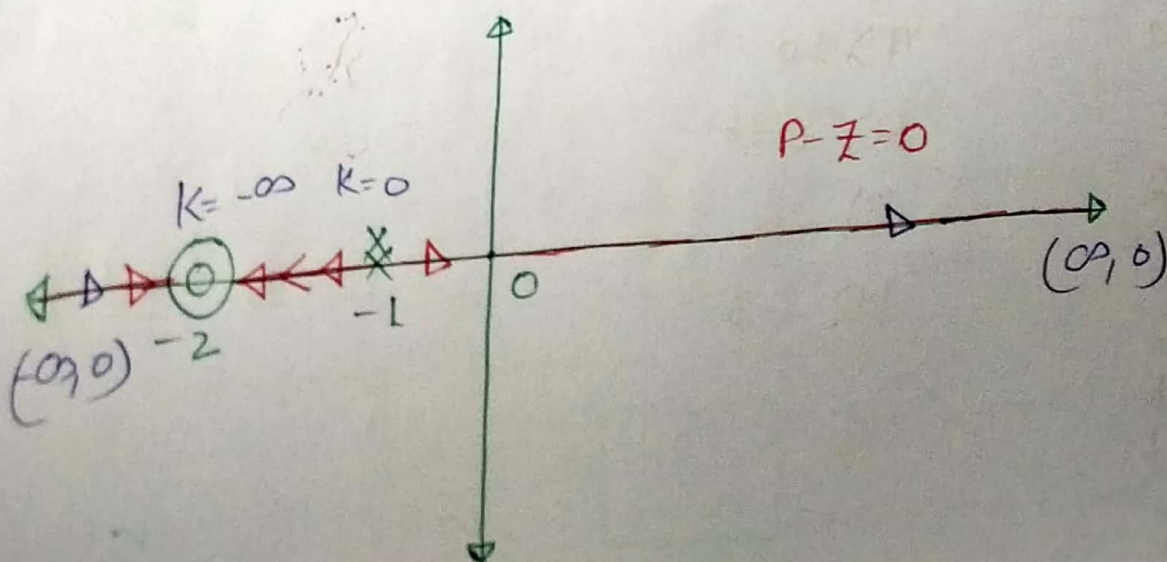
$$\frac{\frac{w}{\sigma + 2} - \frac{w}{\sigma + 1}}{1 + \frac{w^2}{(\sigma + 2)(\sigma + 1)}} = \infty$$

$$\sigma^2 + 3\sigma + 2 + w^2 = 0$$

$$(\sigma + 1.5)^2 + w^2 = (0.5)^2$$

$$C = (-1.5, 0)$$

$$r = 0.5$$



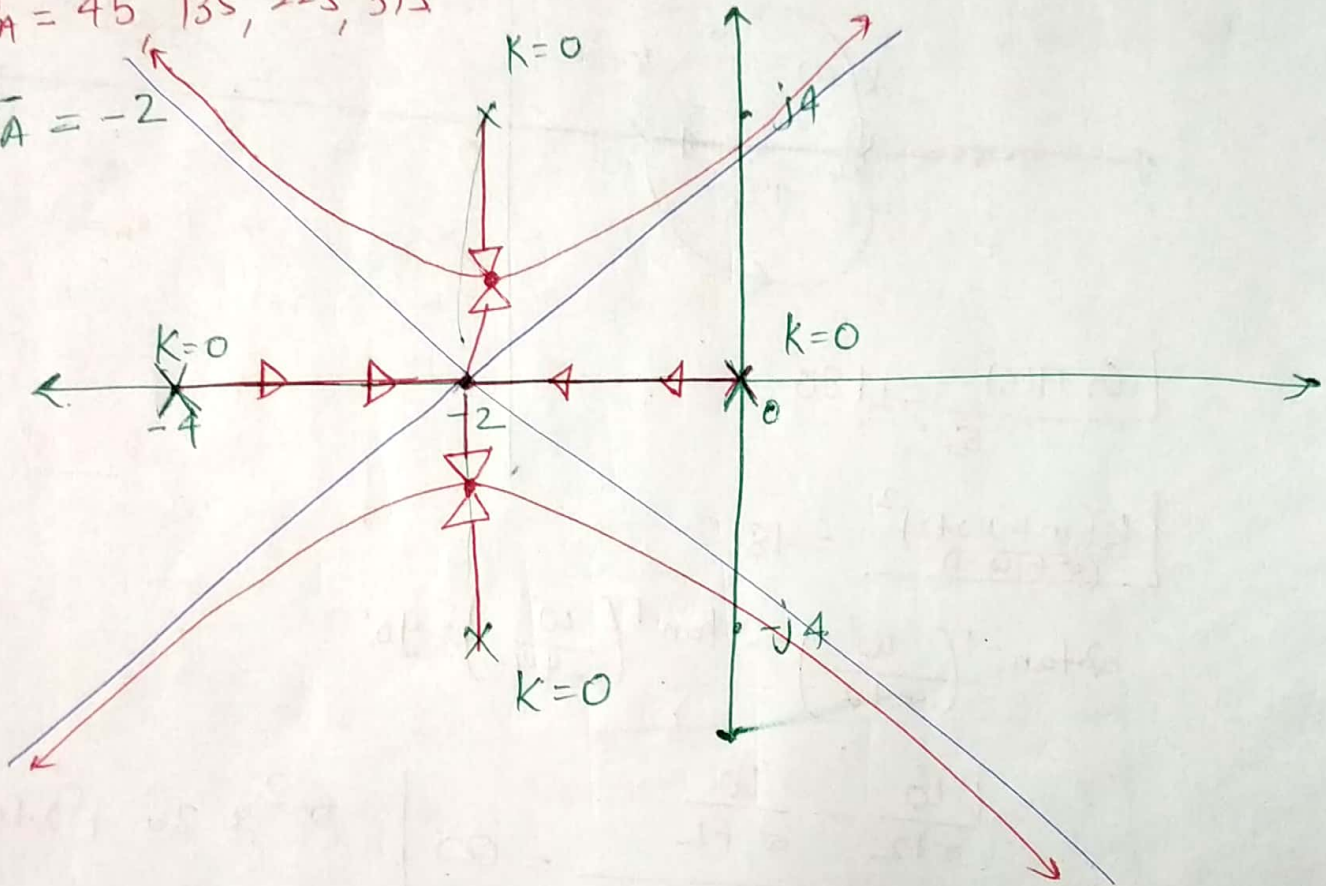
Q. $G H(s) = \frac{k}{s(s+4) \cdot (s^2 + 4s + 20)}$

$\frac{dk}{ds} = -2, -2 \pm j2.45$

$P-Z = 4$

$\phi_A = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

$\sigma_A = -2$



17. b

22. a

23. c

*24. b

25. c

26. c

27. b

28. d

29. c

10. c

11. 10

12. 0.36

13. 25

14. 0.3